

# Applying Language Models to Patient Health Records: Acronym Expansion, Long Document Classification and Explainable Predictions

Aditya M. Kashyap's Dissertation Defense

Jan 20<sup>th</sup>, 2025

Advisers: Chris Callison-Burch, Mary Regina Boland

Committee: Lyle Ungar, Mark Liberman, Li Shen, Ying Ding (University of Texas at Austin)



# Clinical NLP

History of Present Illness: 75 yo male, advanced gastric ca recently started on chemotherapy presenting and hypotension and episode of unresponsiveness after diarrhea and narcotics + megace. Pt was seen on [\*\*] by his oncologist where he was determined to be hypovolemic and received a total of 3 IVF between We and Fri but still c/o poor po intake secondary to abdominal pain and fullness, nausea, and decreased appetite. This morning upon awakening pt was lethargic. He took MSO4, Dilaudid and an unspecified dose of Megace. He then became flushed and pale, had copious diarrhea (non-bloody), then became unresponsive. EMS called and he was found with SBP 80. Narcan given with improvement in mental status but in [\*\*] pt was persistently hypotensive requiring 6 IVF and Levophed. Labs notable for ANC 500 and lactate 17. Sep-sis protocol was initiated. Pt given Flagyl, Levofloxacin, and cefepime and sent to [\*\*].

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1. Abundant use of acronyms and abbreviations.

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This note excerpt contains 282 tokens, but other notes have up to 20,000 tokens!

## Challenges:

1. Abundant use of acronyms and abbreviations.
2. Non-standard use of grammar, punctuation and spelling mistakes.
3. Patients have multiple clinical notes, each spanning between 200 and 20,000 tokens.

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Wealth of Information can be used for research and education!

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# Challenges Addressed in this Dissertation

1. Acronym Disambiguation: **[Kashyap et al., 2020]\***
2. (a) Long Document Classification: **[Kashyap et al., 2023]#**  
(b) Structured and Unstructured Patient Electronic Health Record (EHR) Data classification: **[Kashyap et al., 2023]#**
3. Explainable Inference: **Submitted to Bioinformatics (November 2024)**

\* Kashyap, Aditya, et al. "The CLASSE GATOR (Clinical Acronym SenSE disambiGuATOR): a method for predicting acronym sense from neonatal clinical notes." *International journal of medical informatics* 137 (2020): 104101

# [Kashyap et al., 2023] Kashyap, Aditya, Chris Callison-Burch, and Mary Regina Boland. "A deep learning method to detect opioid prescription and opioid use disorder from electronic health records." *International Journal of Medical Informatics* 171 (2023): 104979

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# Acronym Disambiguation in Clinical Notes

Kashyap, Aditya, et al. "The CLASSE GATOR (Clinical Acronym SenSE disambiGuATOR): a method for predicting acronym sense from neonatal clinical notes." *International journal of medical informatics* 137 (2020): 104101

# Motivation

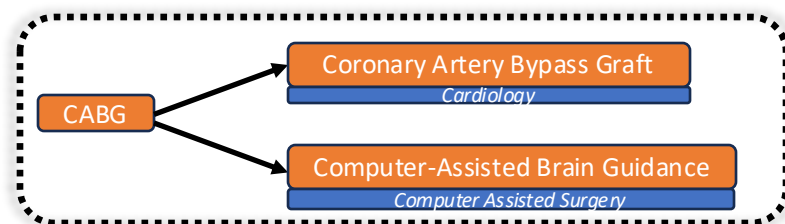
History of Present Illness: 75 **yo** male, advanced gastric **ca** recently started on chemotherapy presenting and hypotension and episode of unresponsiveness after diarrhea and narcotics + megestrol. **Pt** was seen on **Wednesday** by his oncologist where he was determined to be hypovolemic and received a total of 3 **IVF** between **We** and **Fri** but still **c/o** poor **po** intake secondary to abdominal pain and fullness, nausea, and decreased appetite. This morning upon awakening **pt** was lethargic. He took **MSO4**, Dilaudid and an unspecified dose of Megace. He then became flushed and pale, had copious diarrhea (non-bloody), then became unresponsive. **EMS** called and he was found with **SBP** 80. **Narcan** given with improvement in mental status but in **pt** was persistently hypotensive requiring 6 **IVF** and Levophed. Labs notable for **ANC** 500 and lactate 17. Sep-sis protocol was initiated. **Pt** given Flagyl, Levofloxacin, and cefepime and sent to **ICU**.

## 21<sup>st</sup> Century Cures Act

1. Help patients better understand their medical conditions, treatment plans
2. Better equipped to participate in shared-decision making
3. Increased engagement and trust

# What makes this problem hard?

## Variability

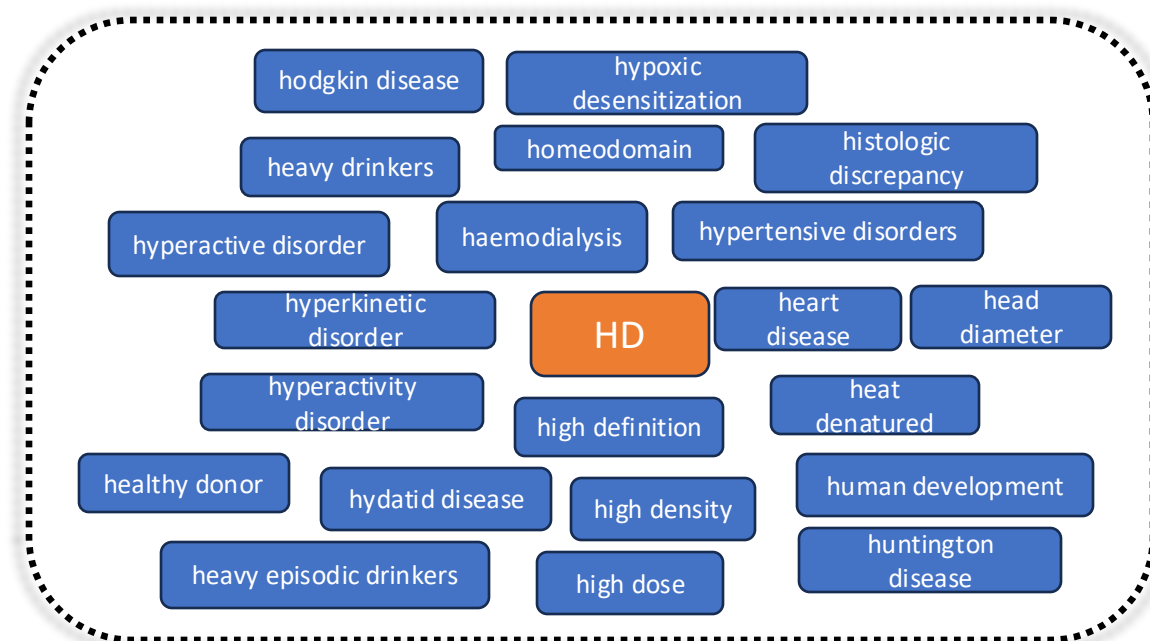


## Complexity and Volume

Patient has Hx of MI, HTN, and DM

*Patient has a history of Myocardial Infarction, Hypertension and Diabetes Mellitus*

## Several Candidate Senses



# Limitations of Past Work

- **Using Rule Based Approaches:**
  - Doesn't transfer well between domains
- **Reliance on Manually Annotated Data:**
  - Expensive: Medical experts required for annotation
    - One study spent over \$100,000 on annotation\*
- **Additional Challenges:**
  - No complete list of clinical abbreviations and their expansions exist
    - invented in an ad hoc manner by health care providers
    - Existing KBs have low coverage
  - Lack of task related public clinical datasets

\*D. Albright, A. Lanfranchi, A. Fredriksen, *et al.*, 2013: "Towards comprehensive syntactic and semantic annotations of the clinical narrative". *Journal of the American Medical Informatics Association : JAMIA*, 20(5), 922–930.

# Approach

1. Data Creation Stage:

2. Model Development Stage:

**Note:** For this task, we focus on the medical domain: ***Prenatal Exposure***

# Approach

## 1. Data Creation Stage:

- Extract acronyms and candidate expansions from over 1M PubMed Research Papers using regular expressions.
- Clean extracted acronym-expansion pairs using a rule-set.
  - 1,257 acronyms and 8,287 expansions.
  - Average number of expansions per acronym is 6.6 (min=2, max=50)
- Extract context information for each acronym-expansion pair using a window.

## 2. Model Development Stage:

| Acronym | Expansion        |
|---------|------------------|
| HS      | High School      |
| HS      | Hepatic Stenosis |
| HS      | Horse Serum      |
| HS      | High Salt        |

**Note:** For this task, we focus on the medical domain: ***Prenatal Exposure***

# Approach

## 1. Data Creation Stage:

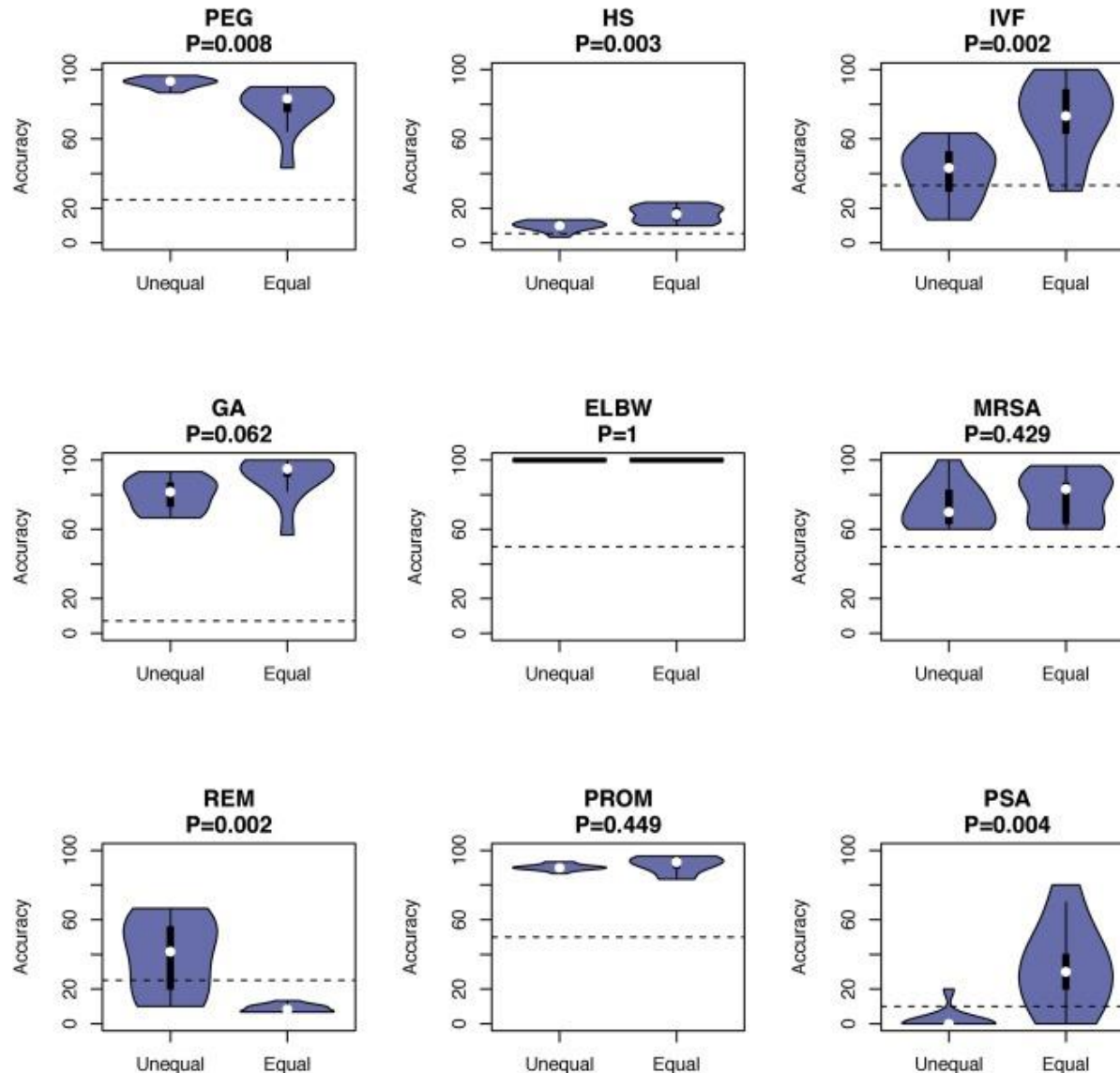
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## 2. Model Development Stage:

- Train Logistic Regression and BioBERT
  - **Input:** acronym + surrounding context
  - **Output:** expansion
- Evaluate on a manually annotated test set of clinical notes

**Note:** For this task, we focus on the medical domain: ***Prenatal Exposure***

# Evaluation on Mimic-III Clinical Notes



Equal vs Unequal class distribution for 9 acronyms on the Mimic-III dataset.  
**Overall accuracy:** 63.04% (dotted line is random acc)  
**Overall random baseline accuracy:** 17.6%

## Reasons for some acronyms performing poorly:

- Some expansions missing from our PubMed dataset (**HS:** Heart sounds)
- Several misspellings ('his', 'has' → hs)
- Domain mismatch between Research text and Clinical Text

## Number of Expansions per acronym:

- **PEG:** 7
- **HS:** 29
- **IVF:** 7
- **GA:** 20
- **ELBW:** 4
- **MRSA:** 2
- **REM:** 5
- **PROM:** 5
- **PSA:** 15

# Contributions and Conclusion

- Created a dataset of 1,257 acronyms and 8,287 expansions with context information
  - Used scientific research papers from PubMed → Previous approaches focused on Wikipedia, Medical webpages (MedlinePlus), and manually annotated clinical notes
- Used BioBERT
  - Past work used ngram features, word embeddings (word2vec) with CNNs, manually created features, Topic Modelling.
- 42.19 % acronyms in our dataset do not occur in UMLS and Medline<sup>1</sup>.
- Cost Effective: Distantly supervised approach removes the need for manually annotated training data.
  - Researchers<sup>2</sup> have spent \$100,000 to annotate 13,091 sentences → \$7.64 per sentence.
  - Moon et al., 2012<sup>3</sup> suggested 500 training annotations required per acronym.
  - Approximate cost to annotate 8,287 acronyms is ~\$32 Million.

1. Wu, Stephen, and Hongfang Liu. "Semantic characteristics of NLP-extracted concepts in clinical notes vs. biomedical literature." *AMIA Annual Symposium Proceedings*. Vol. 2011. American Medical Informatics Association, 2011.

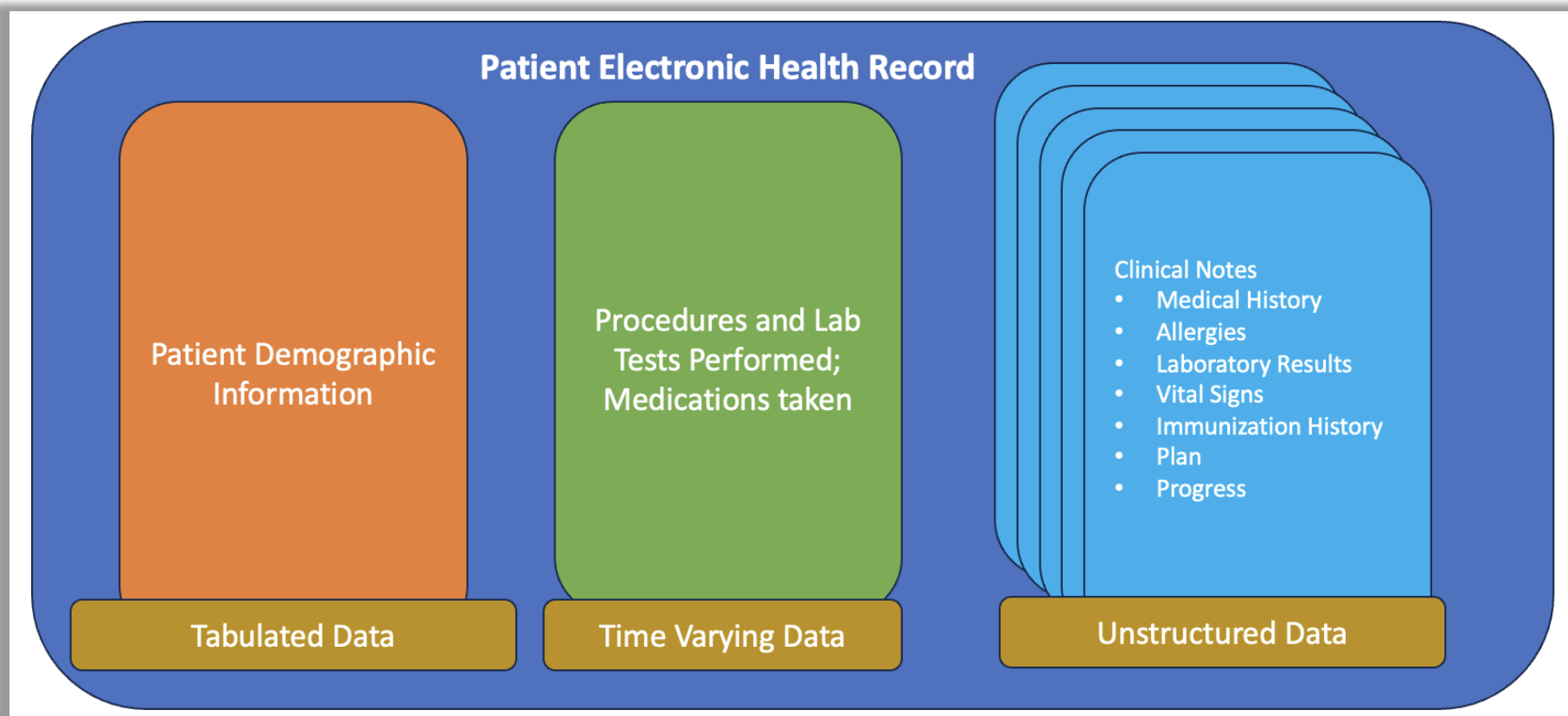
2. D. Albright, A. Lanfranchi, A. Fredriksen, et al., 2013: "Towards comprehensive syntactic and semantic annotations of the clinical narrative". *Journal of the American Medical Informatics Association : JAMIA*, 20(5), 922–930.

3. Moon S, Pakhomov S, Melton GB, 2012 "Automated disambiguation of acronyms and abbreviations in clinical texts: window and training size considerations". *Journal of the American Medical Informatics Association : JAMIA*, 20(5), 922–930.

# Structured and Unstructured Long Document EHR classification

[Kashyap et al., 2023] Kashyap, Aditya, Chris Callison-Burch, and Mary Regina Boland. “A deep learning method to detect opioid prescription and opioid use disorder from electronic health records.” *International Journal of Medical Informatics* 171 (2023): 104979

# Structured and Unstructured Long Document EHR classification



In this section of my dissertation, I build a deep learning model that attempts to solve the following challenges:

- Handle structured and unstructured data
- Handle multiple clinical notes, each containing between 200 and 20000 tokens

**Note:** Patient EHRs also contain images like CT scans, MRI scans, etc. (not shown in the figure)

# Past Research on Long Document Classification

- Bag of Words, TFIDF
  - Disregards word order and context → doesn't capture meaning over long documents
- RNNs and LSTMs
  - Suffer from vanishing gradient problem
  - Difficulty capturing long-term dependencies
- CNNs applied over word embeddings
  - Not as effective at capturing long-term dependencies as attention models
- Transformer Models like BERT
  - Can capture long-term dependencies
  - limited by a maximum input size of 512 tokens
- Hierarchical Attention Models
  - Can handle long documents (> 512 tokens) by analyzing segments and aggregating information across different levels

# Past Research on Long Document Classification

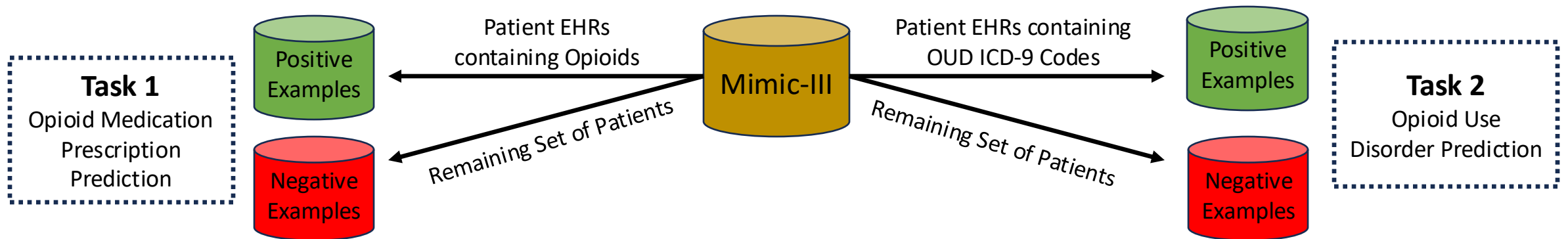
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Inspired by these 2 approaches, we develop a deep learning model that can handle multiple long clinical notes per patient

- Transformer Models like BERT
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- Hierarchical Attention Models
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# Task Formulation and Dataset

- We formulate 2 binary classification problems given patient EHR data:
  1. Will the patient be prescribed an Opioid Medication?
  2. Does the patient have Opioid Use Disorder (OUD)?
- Dataset used:
  - **Mimic III** : De-identified health data associated with ~60,000 CCU admissions



Further Preprocessing:

- Remove opioid medication mentions, or ICD-9 diagnoses codes used to automatically label data.
- Remove or combine sparse demographic features to prevent overfitting.

# Model Development: Example

| Category           | Value                |
|--------------------|----------------------|
| Admission Type     | Emergency            |
| Admission Location | Emergency Room Admit |
| Discharge Location | Home Health Care     |
| Insurance          | Medicare             |
| Language           | English              |
| Religion           | Catholic             |
| Marital Status     | Single               |
| Ethnicity          | White                |

IV Fluid - Normal Saline  
 Blood Glucose Tested  
 Creatinine Tested  
 Urine Output  
 IV Fluid - Lactated Ringer's  
 Sodium Tested  
 Lactate  
 Urine Output  
 IV Fluid - Dextrose 5%  
 White Blood Cell Count Tested  
 Urine Output  
 Naloxone (Narcan) IV  
 Urine Output  
 Urine Output  
 Buprenorphine (Sublingual)

**Patient ID:** 1  
**Admission ID:** 1001  
**Date of Admission:** 2023-01-01  
**Date of Discharge:** 2023-01-02  
**Reason for Admission**  
 The patient was admitted to the hospital for management of opioid overdose following an opioid-related emergency. The patient presented with symptoms of respiratory depression, altered mental status, and signs of acute opioid intoxication. Immediate administration of Naloxone (Narcan) was required to reverse the overdose.  
**Hospital Course**  
 The patient was initially treated in the ICU for opioid overdose and acute renal insufficiency, both of which were complicated by the overdose. The patient was promptly treated with Naloxone (Narcan) (0.4 mg IV), which led to an improvement in the respiratory status and mental clarity. Fluid resuscitation was initiated with IV fluids including Normal Saline and Lactated Ringer's, in addition to Dextrose 5% for supportive care and electrolyte management.  
 The patient's renal function was closely monitored, and creatinine levels were mildly elevated at 1.20 mg/dL, indicating possible kidney involvement due to the opioid use. The patient's sodium levels were stable at 138.5 mmol/L, and blood glucose was within normal range at 95.23 mg/dL.....

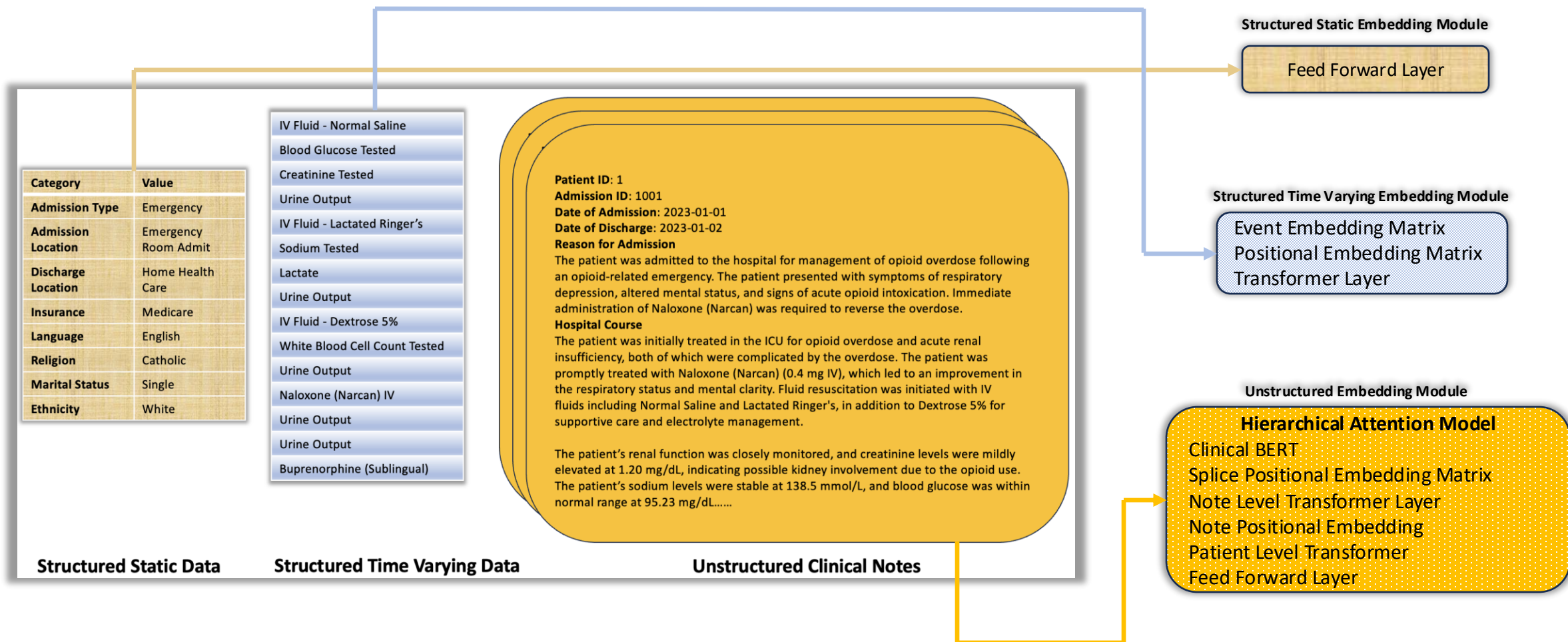
Structured Static Data

Structured Time Varying Data

Unstructured Clinical Notes

*\*This is a synthetic example created for illustrative purposes*

# Model Development: Example



*\*This is a synthetic example created for illustrative purposes*

# Results

## Opioid Prescription Prediction

| Approach              | Test Accuracy    | AUC ROC          |
|-----------------------|------------------|------------------|
| Random Baseline       | 0.5              | 0.5              |
| LogReg ngram baseline | 0.88             | 0.92             |
| Our Approach          | $0.89 \pm 0.003$ | $0.93 \pm 0.002$ |

## Opioid Use Disorder Prediction

| Approach              | Test Accuracy    | AUC ROC          |
|-----------------------|------------------|------------------|
| Random Baseline       | 0.5              | 0.5              |
| LogReg ngram baseline | 0.82             | 0.84             |
| Our Approach          | $0.83 \pm 0.035$ | $0.94 \pm 0.008$ |

**ROC:** Plot between True Positive Rate and False Positive Rate

**AUC ROC:** “Area Under the Curve” of the “Receiving Operating Curve”

- High score suggests model can differentiate well between different classes across different classification thresholds

# Predictive Features for OUD

Used a Global Surrogate LogReg Model to explore most predictive features for OUD

| Structured Static Data   | Structured Time Varying Data | Clinical Notes |
|--------------------------|------------------------------|----------------|
| Age_25-35                | Cryoprecipitate              | methadone      |
| Age_18-25                | Naloxone                     | abuse          |
| Age_35-50                | Acetaminophen-IV             | pain           |
| ADMISSION_TYPE_EMERGENCY | Epinephrine                  | artifact       |
| INSURANCE_Medicaid       | Narcan                       | heroin         |
| GENDER_M                 | Invasive Ventilation         | rhythm normal  |
| ADMISSION_TYPE_URGENT    | Sandostatin                  | overdose       |

According to the literature, age and gender are the most consistent demographic variables in predicting opioid abuse

Similarly, Naloxone, Acetaminophen, Epinephrine, Methadone and Heroin are highly predictive of opioid use disorder

Patients who have a history of substance abuse or who deal with chronic pain have a higher likelihood to develop OUD

# Conclusion

- Contributions of our approach:
  - A Deep Learning model that analyzes EHR data containing structured and unstructured data
  - A Hierarchical Transformer Model + ClinicalBERT that can analyze multiple long clinical notes.
    - Previous approaches in clinical literature: ngram models, word embedding models (GloVe, word2vec) with LSTMs, CNNs
- Limitations:
  - Black box model → Global surrogate models may not reflect the true decision-making process

# Explainable Predictions using Concept Bottlenecks

Submitted to Bioinformatics (November 15<sup>th</sup>, 2024)

# Explainable Predictions using Concept Bottlenecks

- Why is explainability important for clinical NLP and AI?
  - Promotes trust and adoption by healthcare professionals
  - Improve patient safety and mitigates risk
- Potential model improvements:
  - Reduce spurious feature correlations with the predicted output
  - Can help push model towards correct chain of reasoning
  - Reduce gender/racial bias (Not addressed in this thesis but still important)

# Previous Approaches for Explainable Predictions

- **Feature Based Approaches:** Interpreting outputs based on related input features
  - Rationale Extraction: short phrases from the input text (*Lei et al., 2018*)

**Review**

the beer was n't what i expected, and i'm not sure it's "true to style", but i thought it was delicious. **a very pleasant ruby red-amber color** with a relatively brilliant finish, but a limited amount of carbonation, from the look of it. aroma is what i think an amber ale should be - a nice blend of caramel and happiness bound together.

**Ratings**

**Look: 5 stars**      Smell: 4 stars

*\*Image credit: Lei, Tao, Regina Barzilay, and Tommi Jaakkola. "Rationalizing neural predictions.", EMNLP (2016)*

**Figure 1:** An example of a review with ranking in two categories. The rationale for Look prediction is shown in bold.

# Previous Approaches for Explainable Predictions

- **Feature Based Approaches:** Interpreting outputs based on related input features
  - Rationale Extraction: short phrases from the input text (*Lei et al., 2018*)
  - Input Perturbation: Understand predictions based on modifying parts of the input (*Ribeiro et al., 2020*)



(a) Original Image

(b) Explaining *Electric guitar*

(c) Explaining *Acoustic guitar*

(d) Explaining *Labrador*

**Figure 4: Explaining an image classification prediction made by Google's Inception neural network. The top 3 classes predicted are "Electric Guitar" ( $p = 0.32$ ), "Acoustic guitar" ( $p = 0.24$ ) and "Labrador" ( $p = 0.21$ )**

*\*Image credit:* Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin. "Why should i trust you?" Explaining the predictions of any classifier." *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*. 2016.

# Previous Approaches for Explainable Predictions

- **Natural Language Explanation:** Models generate explanations for a given prediction (Camburu et al., 2018)

Premise: A **man** in an orange vest **leans over a pickup truck**.

Hypothesis: A man is **touching** a truck.

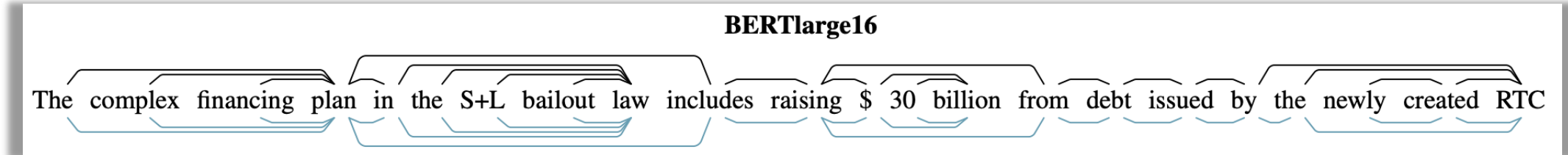
Label: entailment

Explanation: Man leans over a pickup truck implies that he is touching it.

*\*Image credit:* Camburu, Oana-Maria, et al. "e-snli: Natural language inference with natural language explanations." *Advances in Neural Information Processing Systems* 31 (2018).

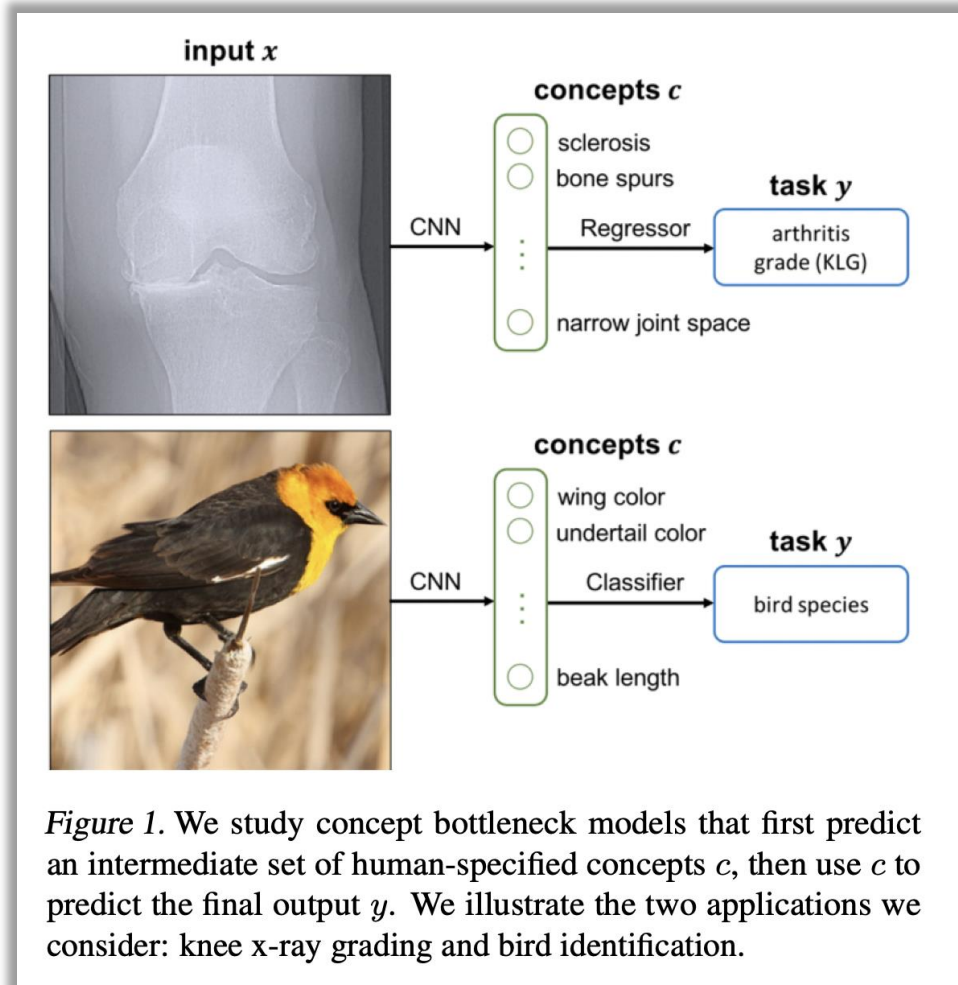
# Previous Approaches for Explainable Predictions

- **Natural Language Explanation:** Models generate explanations for a given prediction (Camburu et al., 2018)
- **Probing Hidden States:** Small classifiers trained on hidden representations for a certain linguistic task (*Hewitt and Manning, 2019*)



*\*Image credit:* Hewitt, John, and Christopher D. Manning. "A structural probe for finding syntax in word representations." *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. 2019.

# Explainable Predictions using Concept Bottlenecks



Yang et al., 2023 → **Language Guided Bottlenecks (LABO)**

- GPT-3 → find bottleneck concepts
- CLIP → Aligned concepts to images
- Linear model → Image classification

In this section of my dissertation, I will focus on using Concept Bottleneck models to build robust, explainable clinical diagnosis models.

*\*Image credit: Koh, Pang Wei, et al. "Concept bottleneck models." International conference on machine learning. PMLR, 2020.*

# Dementia Dataset

Multi-Classification Task: Dementia diagnosis based on patient notes in their EHRs

| <b>Dementia Label</b>       | <b>Number of Patient Admissions</b> |
|-----------------------------|-------------------------------------|
| Vascular Dementia           | 213                                 |
| Alzheimer's                 | 1087                                |
| Other Dementia              | 2650                                |
| No Dementia (Control Group) | 3500                                |

- For each patient admission, we only select the discharge summary note.
- Additionally, we remove phrases such as “dementia”, “alzheimer’s”, “vascular dementia” and other variants from the notes to prevent information leakage.

# How does GPT-4 Perform on Patient Dementia Diagnosis?

## GPT-4 Prompt

For the given patient clinical note, choose one of the following diagnosis labels:

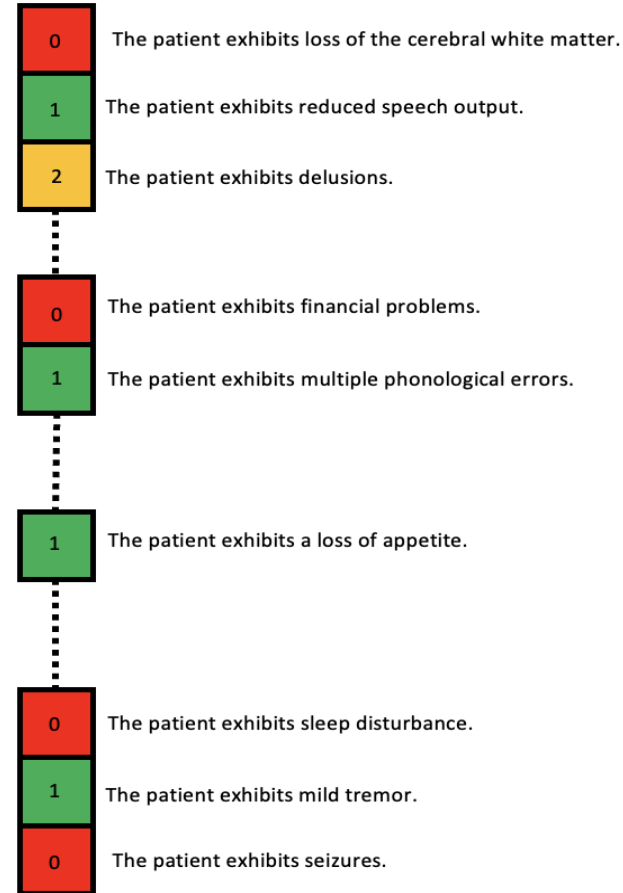
no dementia  
dementia  
alzheimer's dementia  
vascular dementia

You should only reply with the diagnosis label and nothing else.

| Approach          | Dementia Accuracy | Dementia F1 |
|-------------------|-------------------|-------------|
| Random Baseline   | 0.29              | 0.34        |
| Majority Baseline | 0.48              | 0.34        |
| <b>GPT-4</b>      | <b>0.48</b>       | <b>0.40</b> |

# Example of a Concept Vector from a Patient Note

The patient is a [age]-year-old [gender] with a known diagnosis of dementia, presenting with progressive cognitive and behavioral symptoms, including reduced speech output, multiple phonological errors (e.g., phonemic paraphasia), and a significant loss of appetite leading to weight loss. Additionally, the patient might exhibit persistent delusions, such as believing that family members have been replaced by impostors. A mild resting tremor is noted in both hands, though motor coordination remains intact. These findings suggest a worsening of the patient's dementia, possibly with a neurodegenerative etiology such as Alzheimer's or Lewy Body Dementia. Further evaluation, including neurologic assessment and imaging, is recommended to assess the underlying cause of the tremor and cognitive decline.

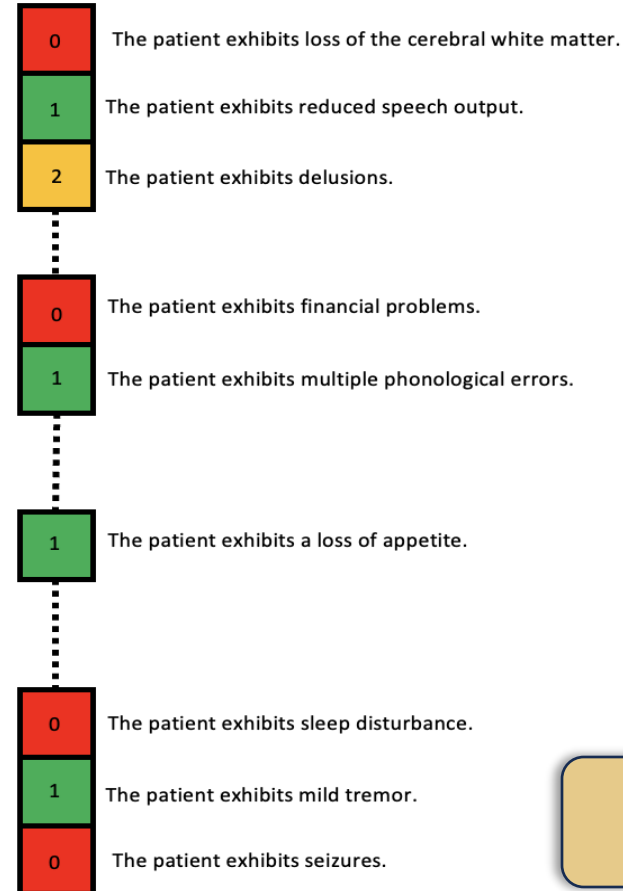


## LEGEND

- 0** Feature is absent
- 1** Feature is present
- 2** Presence of feature is uncertain

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## LEGEND

- 0 Feature is absent
- 1 Feature is present
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How do we obtain this set of concept features?

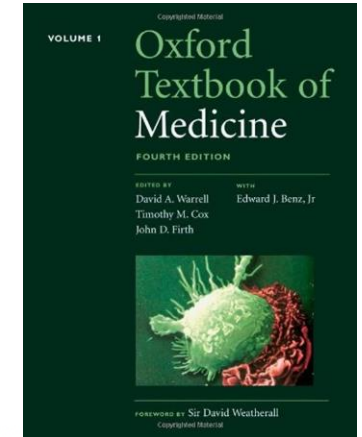
# Obtaining Concepts related to Dementia

- We use the Oxford Textbook of Medicine – 4<sup>th</sup> Edition and GPT-4
  - The chapter on Dementia is first identified
  - GPT-4 is instructed to extract clinical features of dementia
    - Alzheimer's Dementia
    - Vascular Dementia
    - Lewy Body Dementia
    - Frontotemporal Dementia
    - Alcohol related Dementia
    - Pseudodementia
    - Subcortical Dementia

## GPT-4 Prompt:

Rewrite the following description about {condition} as multiple bullet points in the format "The patient exhibits \_\_\_\_". Be as exhaustive as you can.

{textbook description of clinical features}



# Obtaining Concepts related to Dementia

## Dementia with Lewy bodies

### Clinical features

Patients typically present with a progressive cognitive decline paralleling that seen in those with Alzheimer's disease. There are, however, a number of characteristic and distinguishing features. First, there is a tendency to marked spontaneous fluctuations in cognitive abilities, particularly alertness and attention, producing a delirious state lasting days or even weeks. Second, visual hallucinations, illusions, and fleeting misidentification phenomena occur in 50 to 80 per cent of sufferers even at an early stage and without drug provocation. The hallucinations are commonly well-formed images of people or animals. The marked cholinergic deficit is postulated to be the cause of their tendency to visual hallucinations. Third, is the occurrence of spontaneous parkinsonism, which is usually mild in the early stages. Rigidity, gait disturbance, and bradykinesia are all common, although in contrast to patients with Parkinson's disease the tremor is usually mild and atypical (with postural and action components) and symmetrical. Repeated falls also occur. In the later stages the akinetic rigid syndrome can cause severe disabilities in mobility and swallowing an increase in the number of falls. Fourth, there is often an exquisite sensitivity to neuroleptic medication, producing the malignant neuroleptic syndrome (delirium, hyperpyrexia, muscle rigidity, massive elevation of creatine phosphokinase, and renal failure).

- The patient exhibits marked spontaneous fluctuations in cognitive abilities, particularly alertness and attention, producing a delirious state lasting days or even weeks.
- The patient exhibits visual hallucinations, illusions, and fleeting misidentification phenomena.
- The patient exhibits spontaneous parkinsonism, with rigidity, gait disturbance, and bradykinesia.
- The patient exhibits exquisite sensitivity to neuroleptic medication, producing the malignant neuroleptic syndrome.

GPT-4 Output

# Obtaining Concepts related to Dementia

## Dementia with Lewy bodies

### Clinical features

Patients typically present with a progressive cognitive decline paralleling the characteristic and distinguishing features. First, there is a tendency to marked spontaneous fluctuations in cognitive abilities, particularly alertness and attention, producing a delirious state lasting days or even weeks. Second, visual hallucinations, illusions, and fleeting misidentification phenomena occur in 50 to 80 per cent of sufferers even at an early stage and without drug provocation. The hallucinations are commonly well-formed images of people or animals. The marked cholinergic deficit is postulated to be the cause of their tendency to visual hallucinations. Third, is the occurrence of spontaneous parkinsonism, which is usually mild in the early stages. Rigidity, gait disturbance, and bradykinesia are all common, although in contrast to patients with Parkinson's disease the tremor is usually mild and atypical (with postural and action components) and symmetrical. Repeated falls also occur. In the later stages the akinetic rigid syndrome can cause severe disabilities in mobility and swallowing an increase in the number of falls. Fourth, there is often an exquisite sensitivity to neuroleptic medication, producing the malignant neuroleptic syndrome (delirium, hyperpyrexia, muscle rigidity, massive elevation of creatine phosphokinase, and renal failure).

Each sentence extracted by GPT-4 contains multiple clinical features

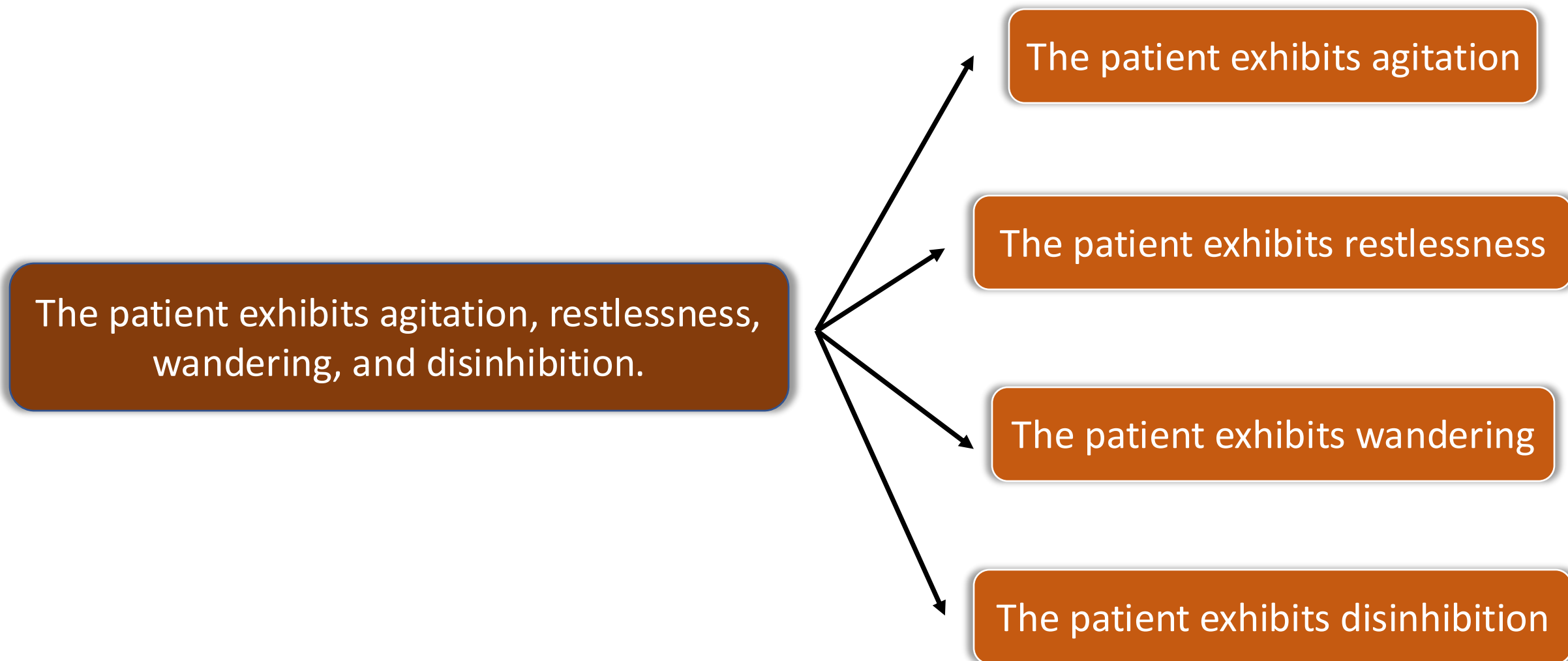
Break them into separate atomic clinical features

- The patient exhibits marked spontaneous fluctuations in cognitive abilities, particularly alertness and attention, producing a delirious state lasting days or even weeks.
- The patient exhibits visual hallucinations, illusions, and fleeting misidentification phenomena.
- The patient exhibits spontaneous parkinsonism, with rigidity, gait disturbance, and bradykinesia.
- The patient exhibits exquisite sensitivity to neuroleptic medication, producing the malignant neuroleptic syndrome.

GPT-4 Output



# Breaking Complex Concepts Into Atomic Concepts



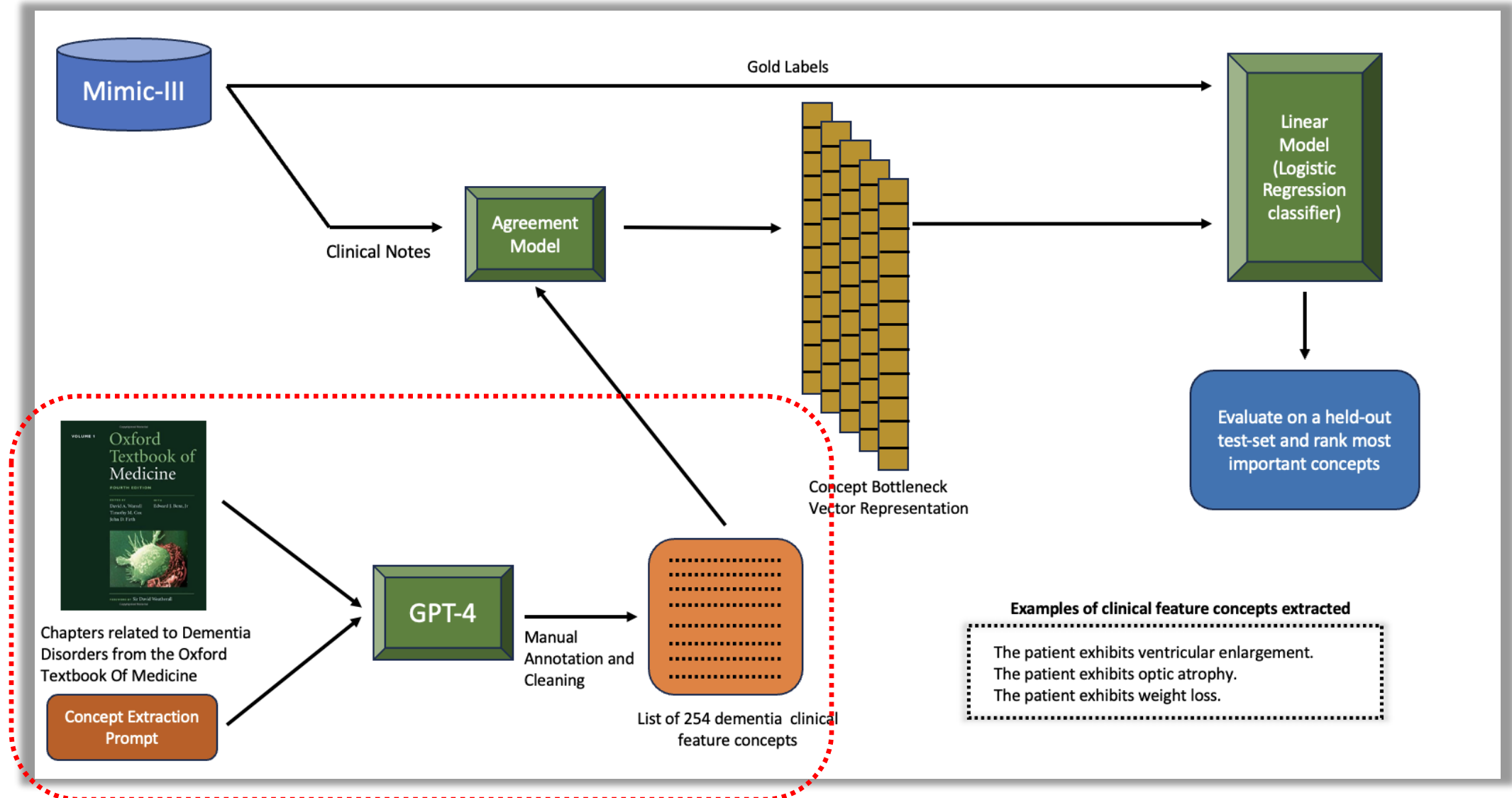
This is achieved using GPT-4 with 2-shot prompting.

# Clinical Features Extracted

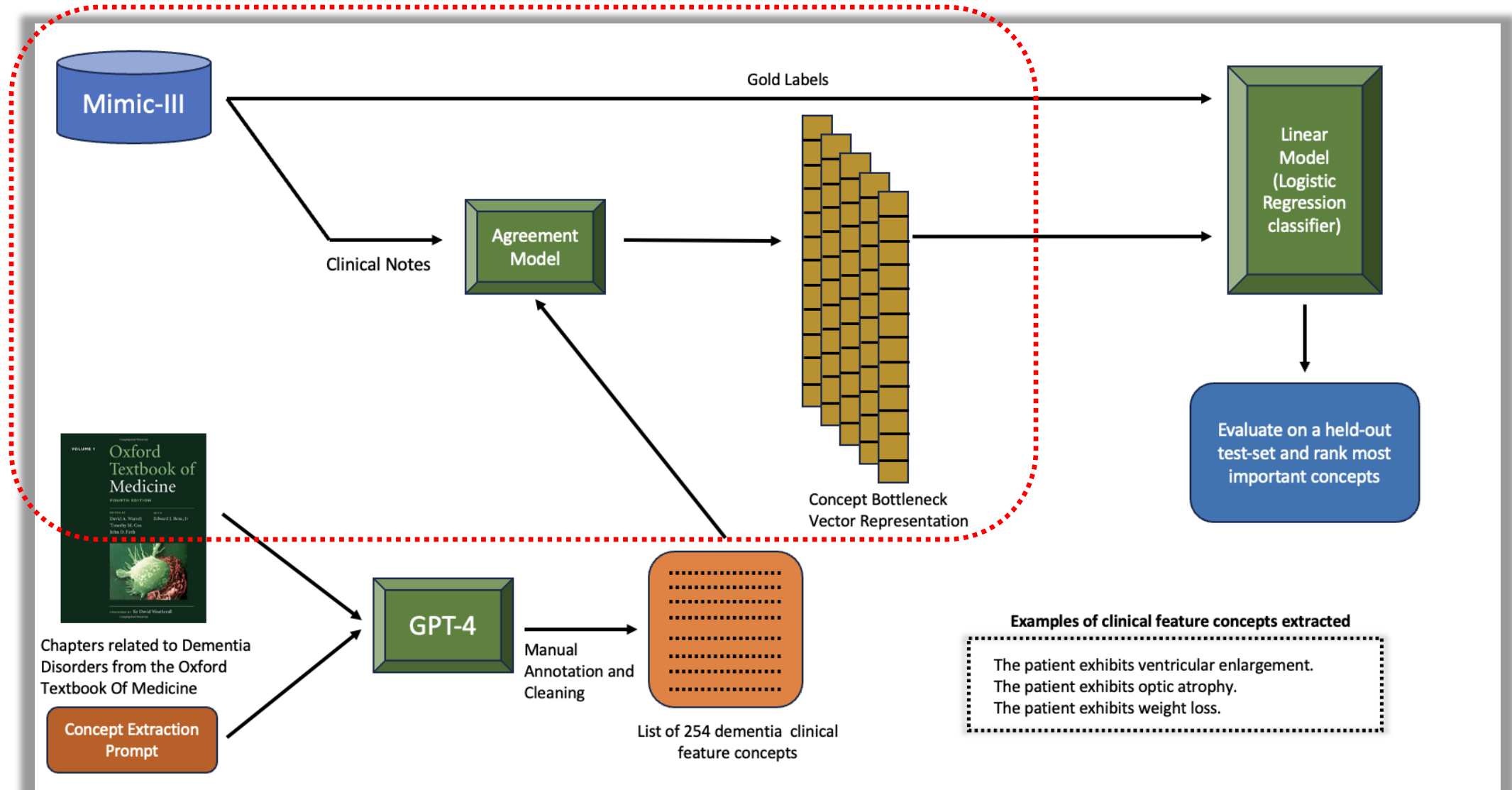
- After manual annotation for quality control, we end up with 254 clinical features.
- Each cell in the table contains a phrase that completes the sentence “The patient exhibits \_\_\_\_.”

| Example Features Extracted from the Oxford Textbook of Medicine |                                             |
|-----------------------------------------------------------------|---------------------------------------------|
| diminished vibration sense                                      | impaired judgement                          |
| difficulty answering questions                                  | neuropsychiatric symptoms                   |
| repetitive questioning                                          | gait apraxia                                |
| large infarcts                                                  | giving bizarrely wrong answers to questions |
| obsessive–compulsive behavior                                   | ventricular enlargement                     |
| optic atrophy                                                   | inconsistency in test performance           |
| dependence                                                      | sleep disturbance                           |
| indifference to domestic responsibilities                       | spontaneous parkinsonism                    |
| increased reflexes                                              | impairment in job performance               |
| greatly increased rigidity                                      | impaired attention                          |

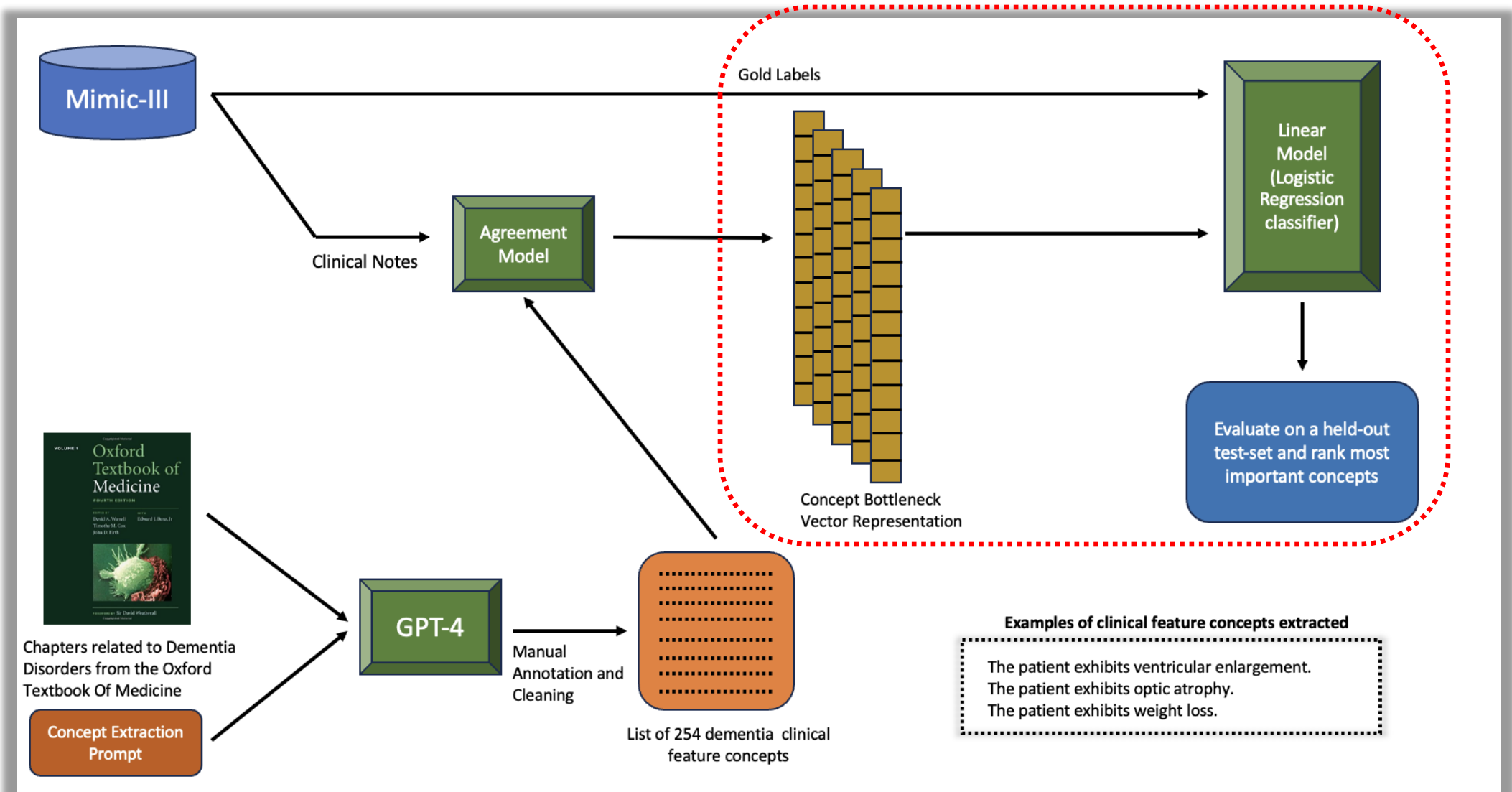
# Pipeline Overview



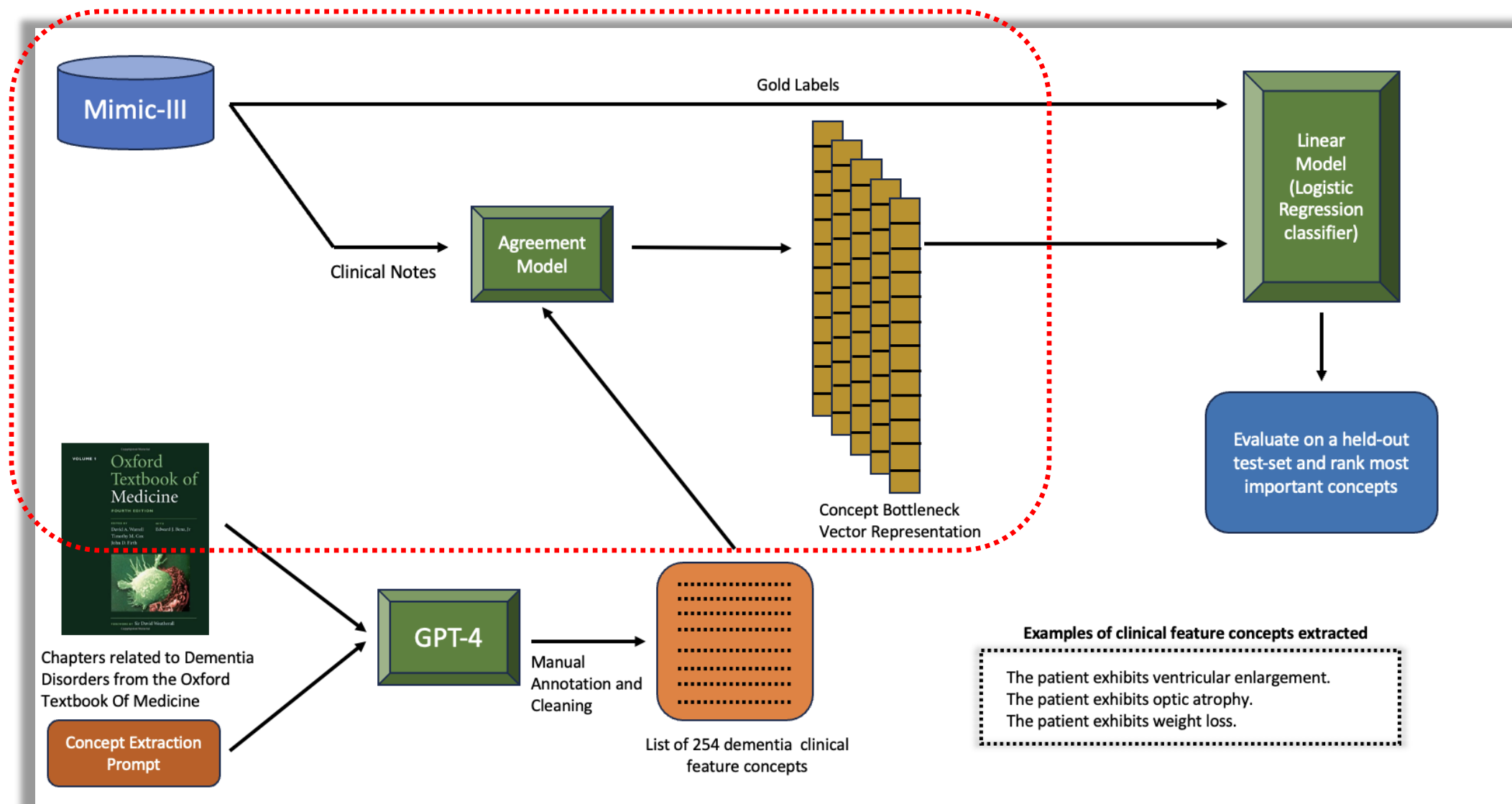
# Pipeline Overview



# Pipeline Overview



# Pipeline Overview



# Agreement Model

- Agreement model predicts concept presence in a clinical note.
  - **Input:** (*Clinical Note, Concept Feature*)
  - **Output:** Either “Yes”, “No” or “Uncertain”
- How do we evaluate the Agreement Model?
  - Manually annotate a subset of 200 (*clinical note, concept feature*) pairs.
    - 10 random *concept features* with 20 *clinical notes* each.

# Agreement Model: Example Input and Output

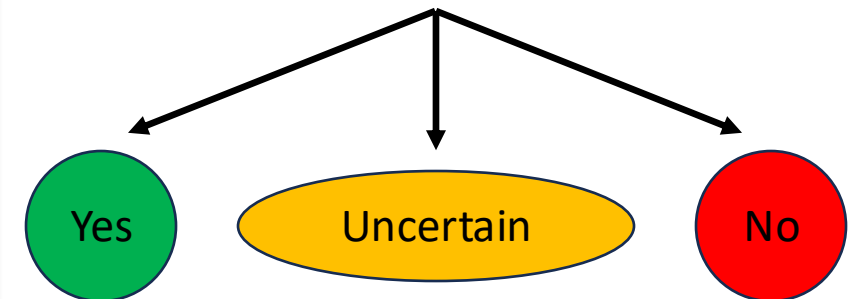
## CLINICAL NOTE

Admission Date: [\*\*2158-1-29\*\*] Discharge Date: [\*\*2158-2-3\*\*] Date of Birth: [\*\*2083-9-9\*\*] Sex: M  
 Service: MEDICINE Allergies: Patient recorded as having No Known Allergies to Drugs  
 Attending:[\*\*First Name3 (LF) 1055\*\*]  
 Chief Complaint: cough and congestion  
 History of Present Illness: 73 year-old man with history of stomach/bladder cancer s/p recent chemo at [\*\*Hospital1 1474\*\*] VA, DM, CAD presentes wit 2-3 days of productive cough with green sputum and sob with exertion, fatigue and lethargy. Per girlfriend, patient had decreased PO intake and mild confusion during that time. Patient with witnessed fall day prior to admission without head trauma. Patient denied dysuria, fevers, frequency. In ER patient noted to have bilateral pneumonia, UTI, hyperglycemia. Patient was started on insulin drip, IVF levaquin and transferred to the ICU for management of hyperosmolar ketoacidosis. No recent med changes.  
 Past Medical History: bladder ca stomach ca DM2 on insulin CAD s/p MI recent admit for chemo tx 6 weeks ago impotence MRSA HTN anemia lacunar CVA CRF glaucoma cognitive decline urinary incontinence, chronic  
 Social History: history of alcohol abuse in past no drugs, no smoking  
 Physical Exam:  
 VS: temp: 102.4 bp: 170/91 HR: 93 RR: 20 99% rm air, general: somnolent but AAOx3, HEENT: MMM, no JVD, no Virchow's node, no nuchal rigidity, neuro: CNII-XII intact, [\*\*6-1\*\*] strength in upper extremities, DTR's intact  
 Pertinent Results: Admit labs: [\*\*2158-1-29\*\*] 12:12AM WBC-9.5 RBC-2.85\* HGB-8.5\* HCT-27.3\* MCV-96  
 .....  
 Followup Instructions:  
 Please follow-up with your primary care physician [\*\*Last Name (NamePattern4) \*\*] [\*\*1-29\*\*] weeks.

## CONCEPT FEATURE

Does the patient exhibit confusion?

## OUTPUT



# Agreement Model: Example Input and Output

## CLINICAL NOTE

Admission Date: [\*\*2158-1-29\*\*] Discharge Date: [\*\*2158-2-3\*\*] Date of Birth: [\*\*2083-9-9\*\*] Sex: M  
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 .....  
 Followup Instructions:  
 Please follow-up with your primary care physician [\*\*Last Name (NamePattern4) \*\*] [\*\*1-29\*\*] weeks.

## CONCEPT FEATURE

Does the patient exhibit confusion?

OUTPUT

Yes

# Agreement Model Approaches

- Three different agreement model approaches:
  - GPT-4 (Closed Source)
  - Llama 2 (Open Source)
  - GPT-4 with Text Embeddings (Closed Source)

# Input for GPT-4 as Agreement Model

## System Prompt

You are a clinical research assistant. For the given clinical note, answer the question ("Concept Question") provided in the dictionary. Use "Definition", "Possible Responses", "Response Guide" and "Additional Response" to help better answer the question. Your response should be of the following format:

```
{"Response": Either 'Yes', 'No' or 'Uncertain',  
  "Supporting Information": If "Response" is either "Yes" or "Uncertain", provide supporting sentences from the  
  clinical note. If "Response" is 'No', this should be "None"}
```

## Chat History

Clinical Note

## User Message

Feature Name: <>  
Concept Question: <>  
Definition: <>  
Possible Responses: <>  
Response Guide: <>

# Input for GPT-4 as Agreement Model

**"Feature Name":** "Binswanger's disease",

**"Concept Question":** "Can the following sentence be inferred from the clinical note?: 'The patient exhibits Binswanger's disease.'",

**"Definition":** "Binswanger's disease, also known as subcortical vascular dementia, in a clinical setting refers to a form of dementia caused by damage to the white matter of the brain resulting from chronic, small blood vessel disease. The damage leads to progressive memory loss, decline in cognitive skills, and changes in mood or behavior. It is often associated with hypertension, diabetes, and other risk factors for vascular disease. Symptoms can also include slow, unsteady gait, urinary symptoms, and mood changes. Although the disease is progressive, managing the underlying vascular conditions can help slow its progress.",

**"Possible Responses":** "Yes, No, Uncertain",

**"Response Guide":** "Yes: The clinical note explicitly mentions that the patient has the clinical feature.\nNo: The clinical note does not explicitly mention that the patient has the clinical feature and there is also no information that might suggest the patient has the clinical feature.\nUncertain: The clinical note contains some information that might suggest the patient has the clinical feature (through correlations), but it is not explicitly stated."

# GPT-4 as Agreement Model: Results

- Dementia dataset: 7,450 patient notes + 254 clinical features.
- 1.9 million (7,450 x 254) *patient note – concept feature* pairs containing 7.9 billion tokens.

| Agreement Model   | Concept Accuracy | Concept F1  |
|-------------------|------------------|-------------|
| Random Baseline   | 0.31             | 0.35        |
| Majority Baseline | 0.65             | 0.51        |
| <b>GPT-4</b>      | <b>0.86</b>      | <b>0.84</b> |

Technical Challenges of using GPT-4 to annotate all pairs:

- **Time:** It took over 40 days for GPT-4 to annotate all pairs!
- **Cost:** The cost of using the synchronous GPT-4 API as of June 2024 for our dataset was \$240,540.
  - 7.9 billion input tokens (\$30/1M input tokens)
  - 59 million output tokens (\$60/1M output tokens)

More efficient agreement models?

- Open-Source Model: Llama2
- GPT4 with Text Embeddings

# Llama-2 as Agreement Model

- Although GPT-4 performs well across different tasks, high costs and limited accessibility makes it less attainable for individuals and smaller organizations.
- Finetune Llama-2 using Knowledge Distillation
  - First, GPT-4 is used to create a training set of 125,000 (*patient note, clinical feature*) pairs with *Yes, Uncertain* and *No* labels.
  - We finetune a 7B and 13B Llama2 model using LoRA, 4-bit precision and gradient accumulation.

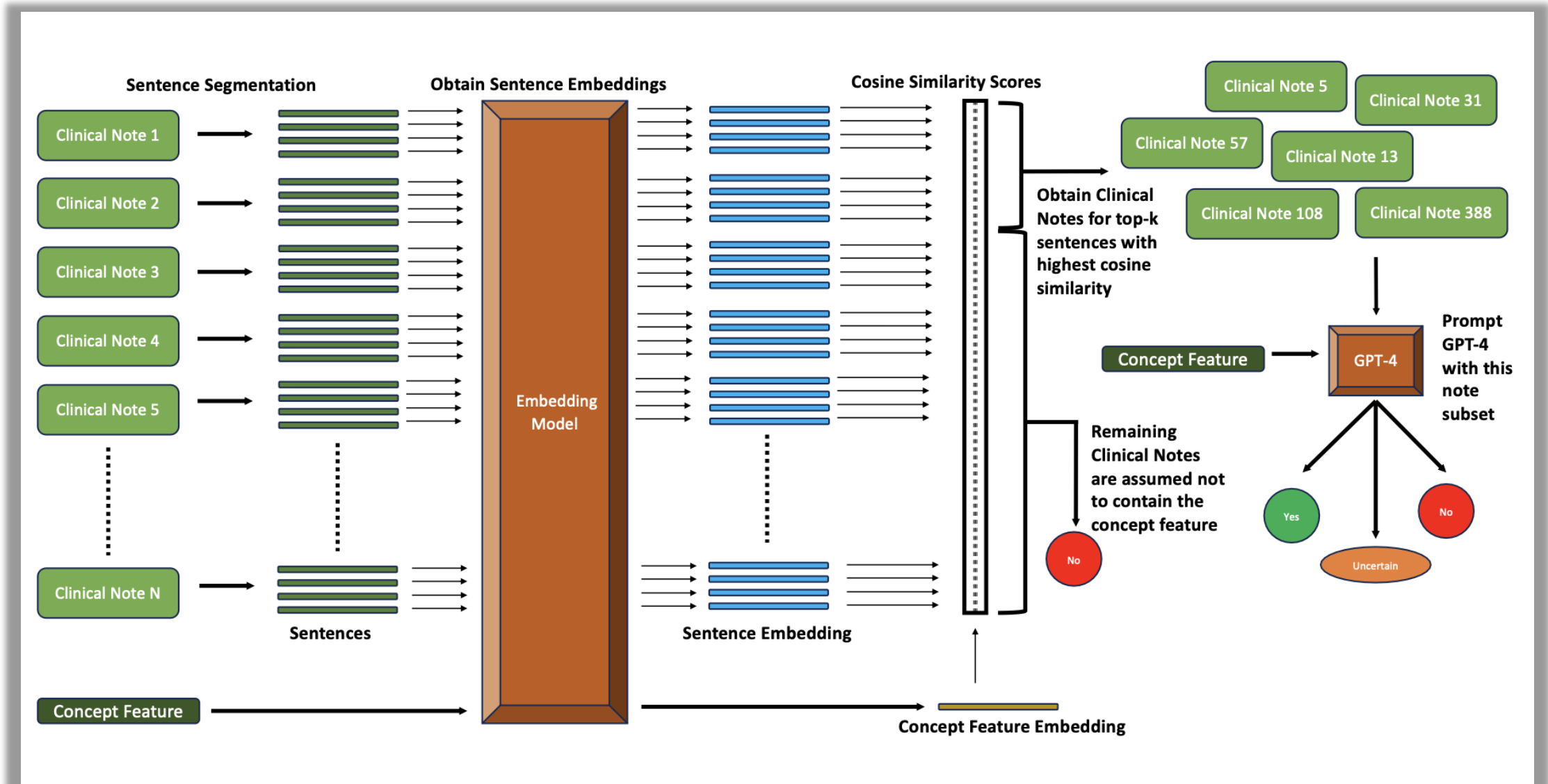
| Agreement Model     | Concept Accuracy | Concept F1  |
|---------------------|------------------|-------------|
| Random Baseline     | 0.31             | 0.35        |
| Majority Baseline   | 0.65             | 0.51        |
| GPT-4               | 0.86             | 0.84        |
| <b>Llama2 – 7B</b>  | <b>0.43</b>      | <b>0.38</b> |
| <b>Llama2 – 13B</b> | <b>0.47</b>      | <b>0.50</b> |

Can we use the GPT-4 agreement model more efficiently?

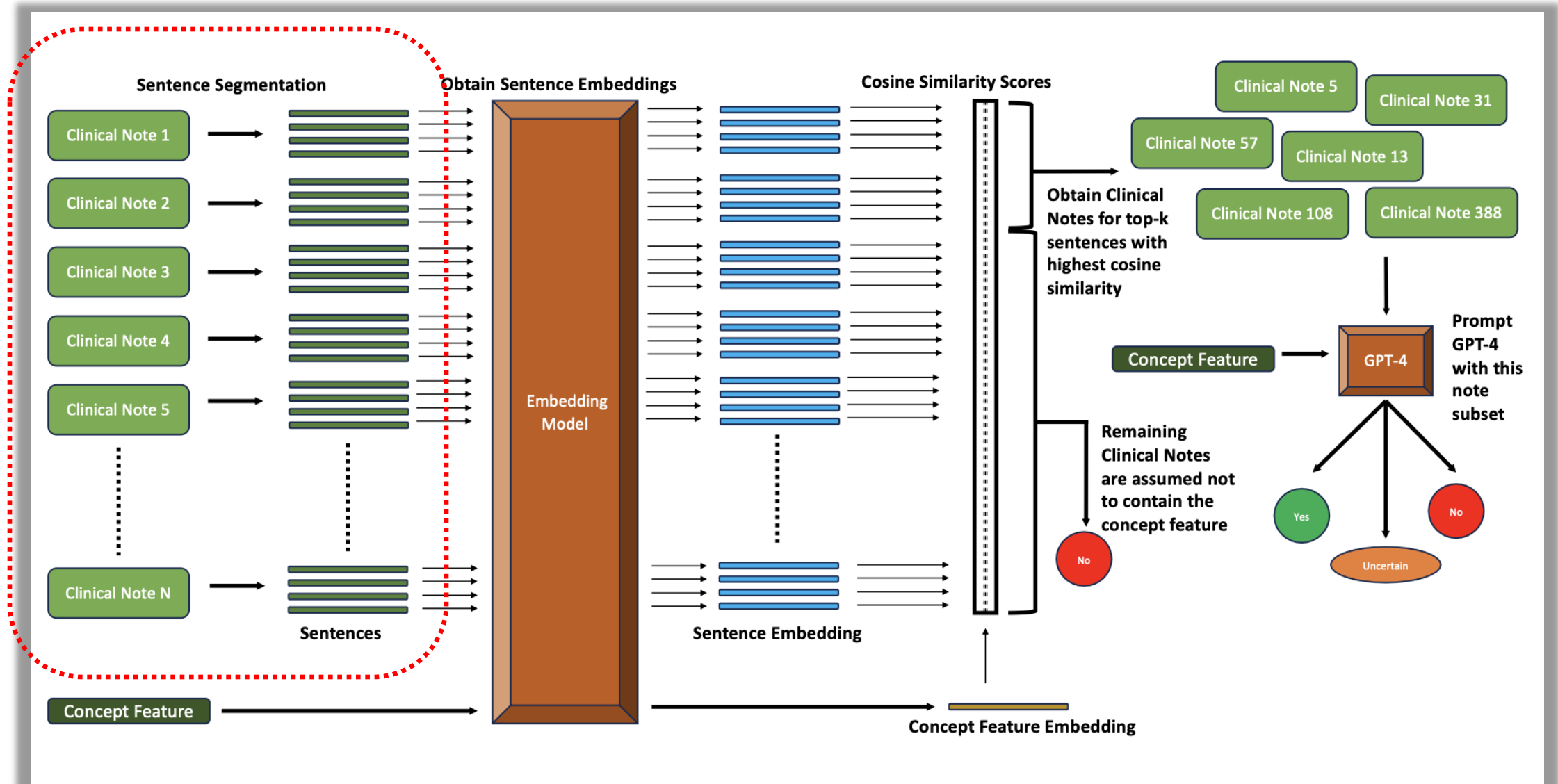
# GPT-4 with Text Embeddings as Agreement Model

- Reduce number of (*patient note*, *concept feature*) pairs that GPT-4 must label.
- Some concepts are less common than others, occurring in only a fraction of the patient notes.
- Text Embeddings → identify pairs where the *patient notes* are likely to have the corresponding *concept feature* activated.

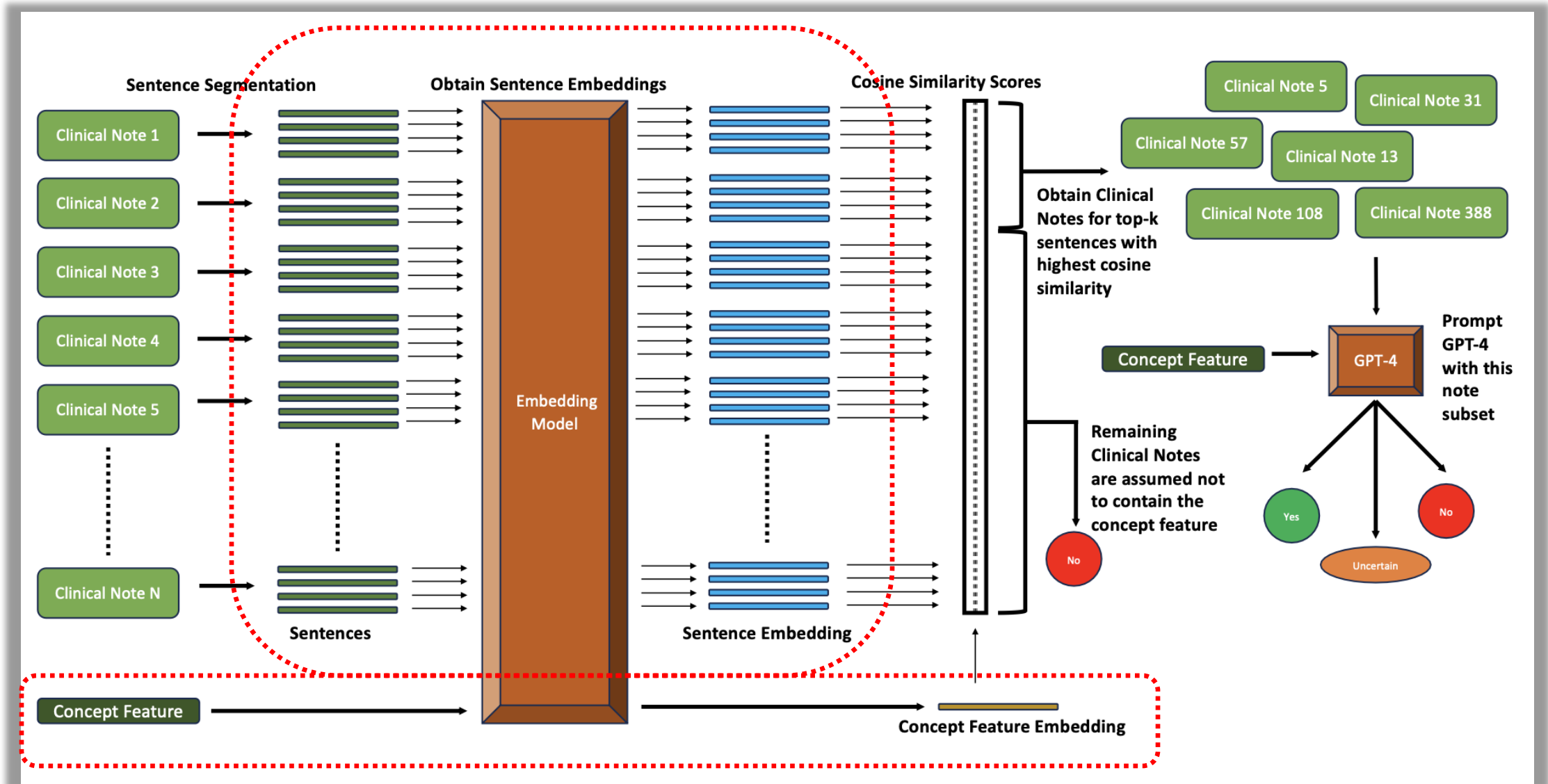
# GPT-4 with Text Embeddings as Agreement Model: Pipeline Overview



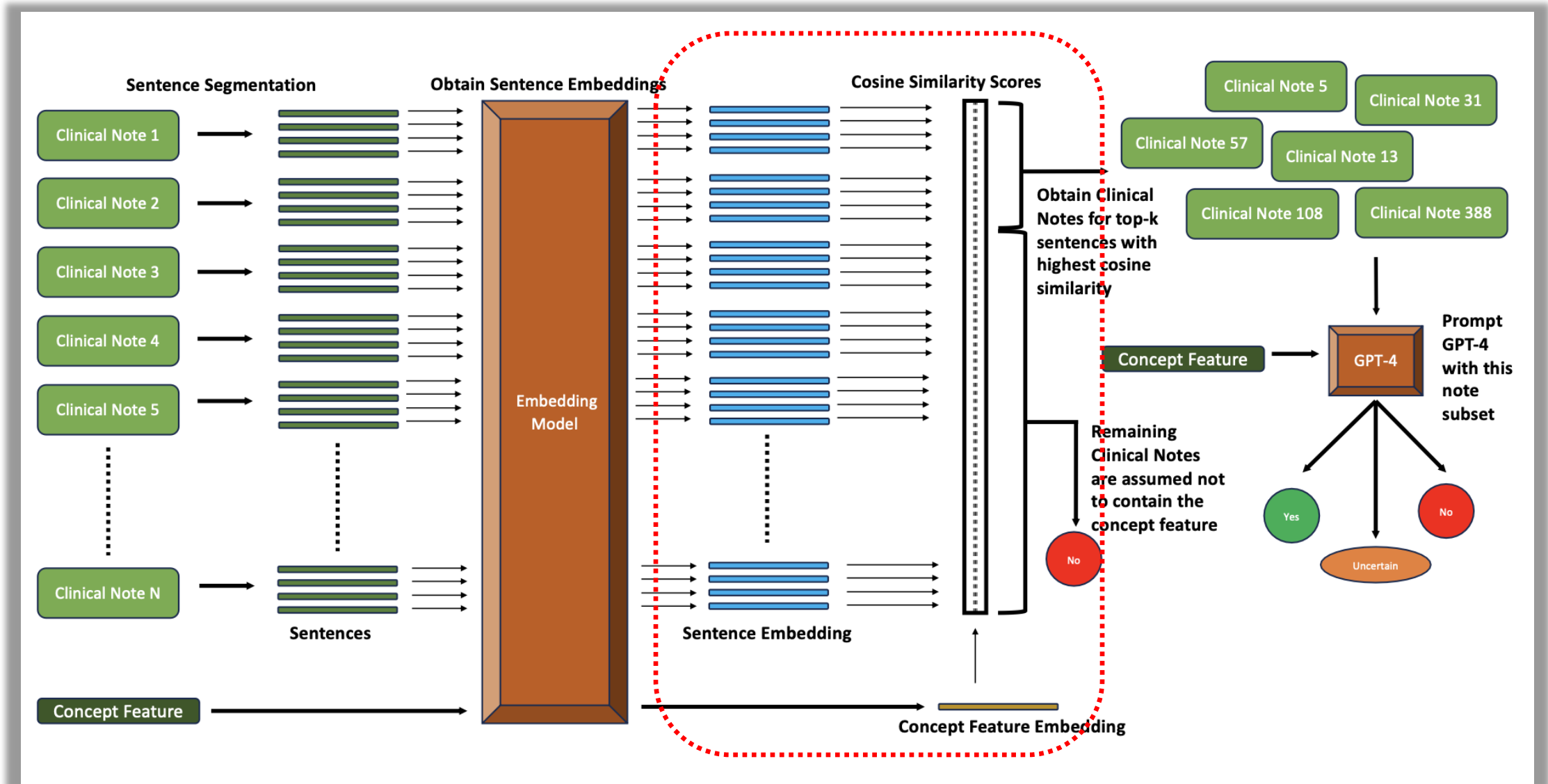
# GPT-4 with Text Embeddings as Agreement Model: Pipeline Overview



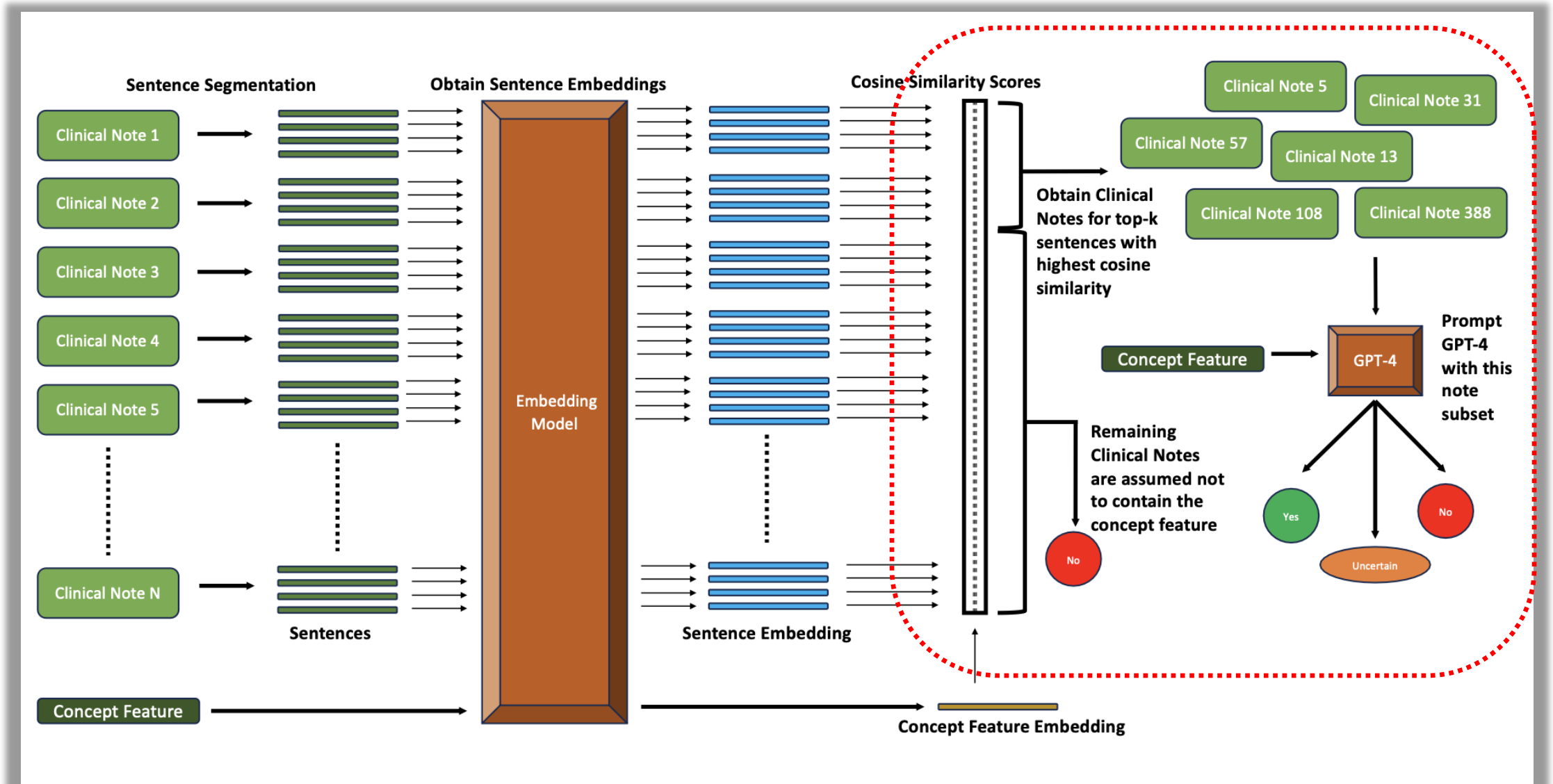
# GPT-4 with Text Embeddings as Agreement Model: Pipeline Overview



# GPT-4 with Text Embeddings as Agreement Model: Pipeline Overview



# GPT-4 with Text Embeddings as Agreement Model: Pipeline Overview



# GPT-4 with Text Embeddings as Agreement Model

- **Clinical Note Segmentation:**
  - 1,2,3,4,5-sentence chunks of the clinical notes.
  - *2-sentence chunks* work best.
- **Text Embedding Model:**
  - OpenAI models (text-embedding-ada-002, text-embedding-3-small, text-embedding-3-large) and INSTRUCTOR models.
  - *text-embedding-3-large* works best.
- **Top-k Sampling:** Extrapolation
  - Randomly select 300 patient notes from our dementia dataset
  - Use GPT-4 to annotate the presence of each of the 254 concept features.
  - Based on concept-frequency in the 300-note subset, extrapolate to 7,450 notes
    - For example, if the concept feature “The patient exhibits multiple lesions” occurs 60 times in the 300-note subset, then roughly 1,500 patient notes will contain it in the entire 7,450 dementia dataset.

# GPT-4 with Text Embeddings as Agreement Model: Results

While the GPT-4 + text embedding approach shows slightly lower performance, the following advantages are seen:

- Total time required was 2 days (compared to over 40 days without text embeddings)
- Total cost was \$7,276 (compared to \$240,540 without text embeddings)

| Agreement Model                | Concept Accuracy | Concept F1  |
|--------------------------------|------------------|-------------|
| Random Baseline                | 0.31             | 0.35        |
| Majority Baseline              | 0.65             | 0.51        |
| GPT-4                          | 0.86             | 0.84        |
| <b>GPT-4 + Text Embeddings</b> | <b>0.82</b>      | <b>0.78</b> |

This approach makes it more efficient to scale up to larger datasets and other clinical outcomes while keeping cost and runtime low.

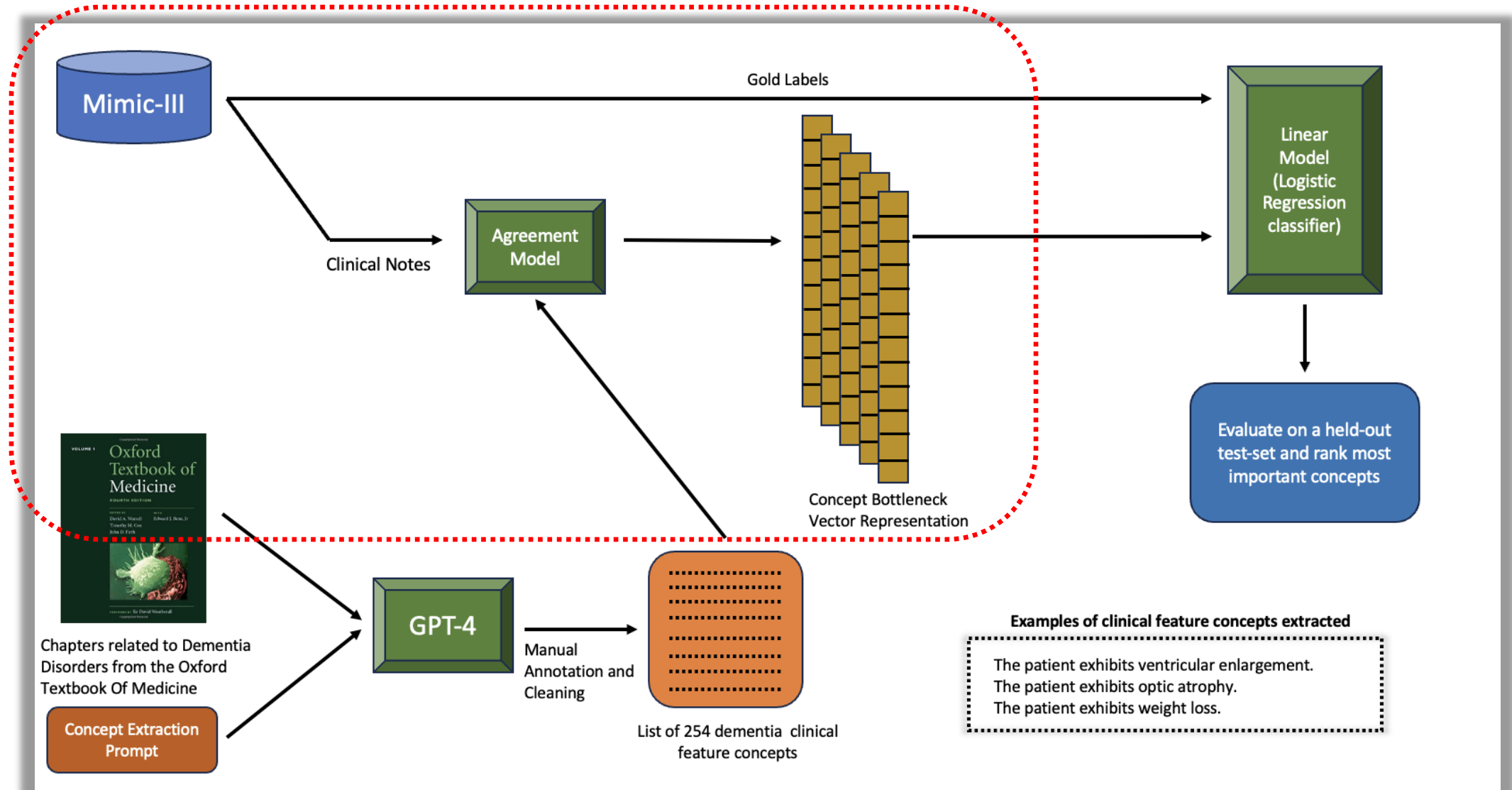
## Additional Advantage of using Text Embeddings with GPT-4

With our Text Embedding + GPT-4 approach, we can additionally find note sentences that correspond to an activated concept.

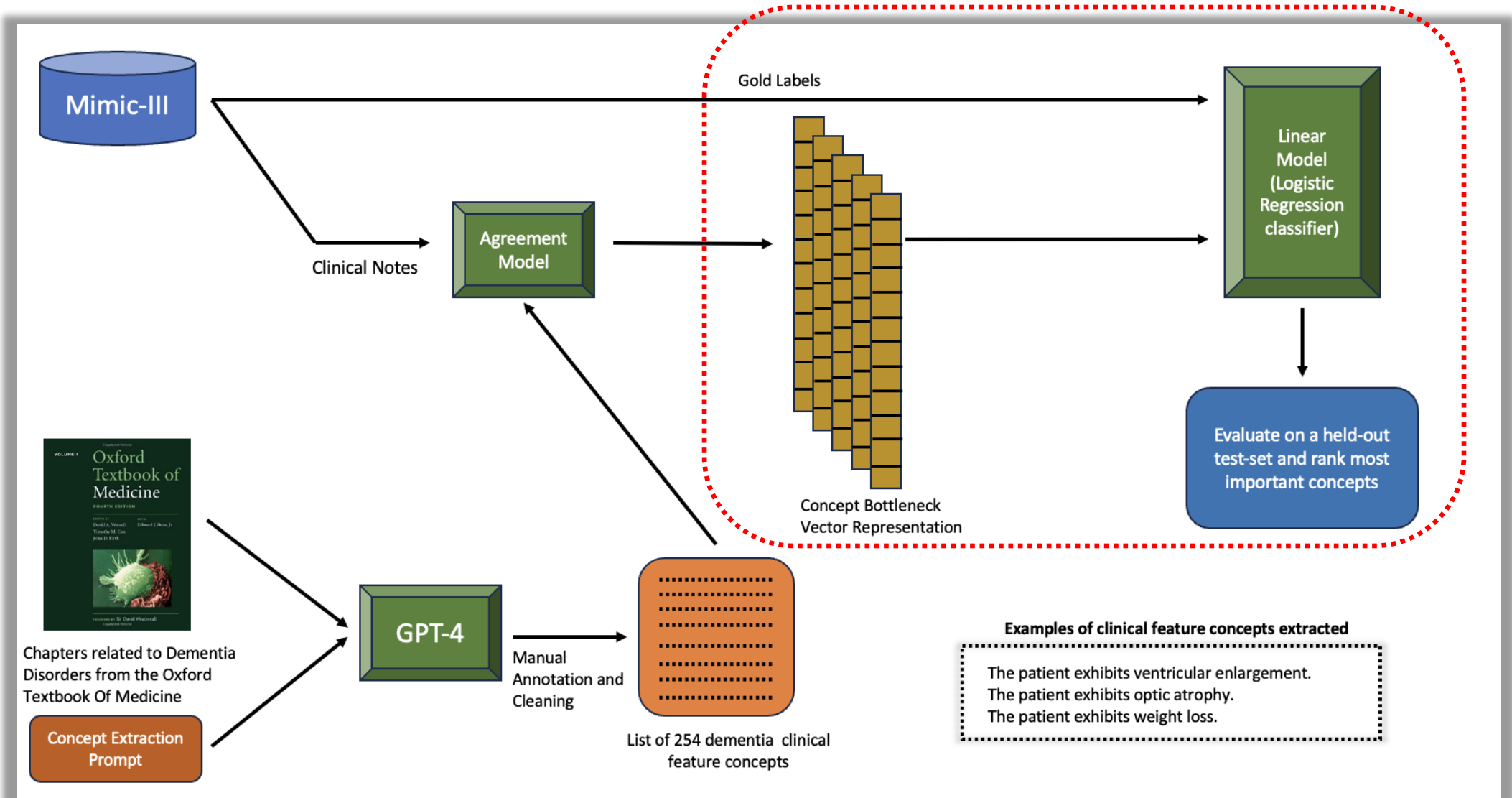
This provides an additional layer of transparency that makes it easier to control for quality.

| Concept Feature                                       | Sentence in Clinical Note                                                                                                |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| infarcts involving multiple main arterial territories | Head CT showed multiple old infarcts but no new lesion.                                                                  |
|                                                       | Past Medical History: Right temporal \& left parietal infarcts                                                           |
| history of atherosclerotic risk factors               | Past Medical History: Idiopathic peripheral neuropathy (per family), HTN...                                              |
|                                                       | Past Medical History: - hypertension. Social History: She is a former smoker                                             |
| decreased reflexes                                    | Her reflexes revealed she was hyporeflexic in the lower extremities and hyperreflexic in the upper extremities.          |
|                                                       | Reflexes: Brisk in the upper extremities, normal at patellae and reduced at achilles. The plantars are mute bilaterally. |

# Pipeline Overview



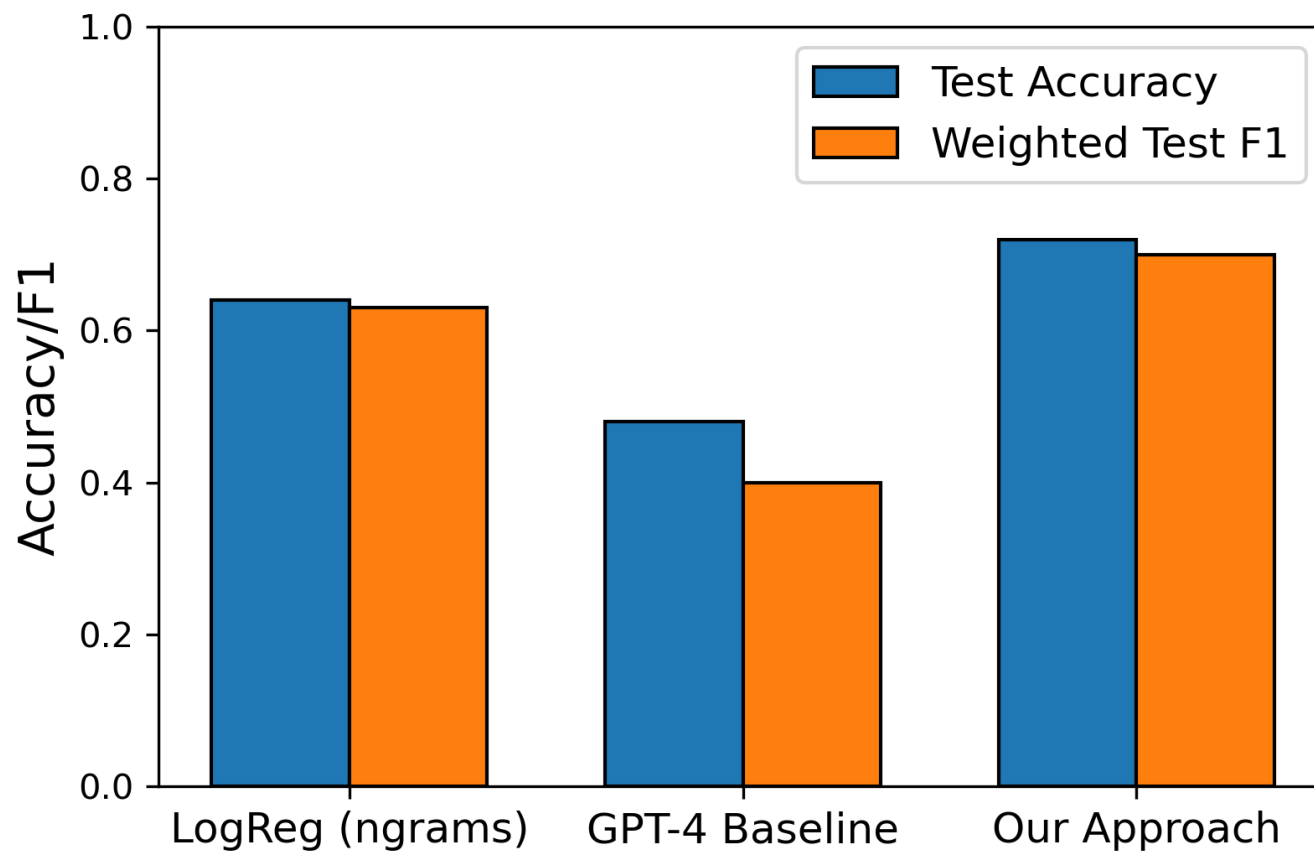
# Pipeline Overview



# Dementia Prediction using Patient Concept Vectors

- Every patient note → Concept Vector using the Agreement Model.
  - each vector index corresponds to one of 254 unique concepts features
- We divide the 7,450 notes in our dementia into train-val-test (0.90/0.05/0.05)
- Finally, we train a linear model (Logistic Regression) to predict dementia type.
- We also compare our approach to 2 baselines: GPT-4 and ngram-logistic regression.

# Dementia Prediction: Results



Our approach achieves a higher accuracy and F1 score compared to the baselines.

Our approach also utilizes high level clinical concepts to predict dementia type.

# Top Predictive Features of our Concept Bottleneck Approach

| No Dementia                                                | Other Dementia                                             | Alzheimer's Dementia                                               | Vascular Dementia                                                  |
|------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|
| progressive cognitive decline                              | poor performance on semantically based tasks               | memory loss that may be less prominent than in Alzheimer's disease | memory loss that may be less prominent than in Alzheimer's disease |
| cognitive impairment                                       | impaired comprehension of words                            | decreased reflexes                                                 | narrowing of the vessel                                            |
| poor memory                                                | amnesia                                                    | profound loss in conceptual knowledge (or semantic memory)         | infarcts involving multiple main arterial territories              |
| poor concentration                                         | myoclonus                                                  | rapid onset of memory impairment                                   | poor performance on semantically based tasks                       |
| history of atherosclerotic risk factors                    | bladder dysfunction                                        | mental inflexibility                                               | history of atherosclerotic risk factors                            |
| rigidity                                                   | poor concentration                                         | progressive cognitive decline                                      | forgetfulness                                                      |
| profound loss in conceptual knowledge (or semantic memory) | profound loss in conceptual knowledge (or semantic memory) | fluent speech                                                      | apathy                                                             |
| amnesia                                                    | large infarcts                                             | visuospatial difficulties                                          | poor performance on neuropsychological tests                       |
| substance dependence                                       | reduced speech output                                      | lack of initiation                                                 | large infarcts                                                     |
| subcortical dementia syndrome                              | fluent speech                                              | severe forgetfulness                                               | restlessness                                                       |

Each cell in the table contains a concept of the form “The patient exhibits \_\_\_\_.”

## LEGEND



Feature is absent



Feature is present



Presence of feature is uncertain

# Exploring Gradient Tree Boosting as Final Classification Model

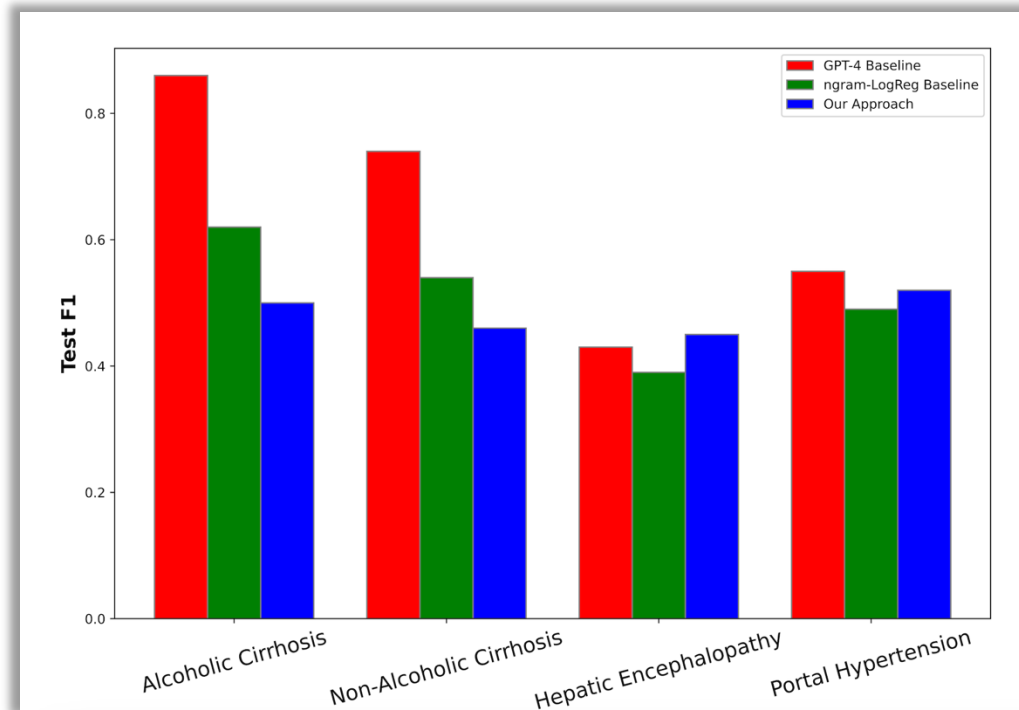
- Disadvantage of Logistic Regression as the final classification layer:
  - Cannot handle interacting features automatically
- Gradient Tree Boosting as final layer:
  - Can automatically handle interacting features (up-to a maximum depth of the Decision Tree weak learner used)

| Final Classifier       | Test Accuracy | Test F1 |
|------------------------|---------------|---------|
| Logistic Regression    | 0.71          | 0.70    |
| Gradient Tree Boosting | 0.72          | 0.71    |

# Liver Disorder Prediction

- How well does our approach work for other Clinical Outcomes?
- Choose liver disorders and rerun the entire pipeline:

| Liver Disorder Type                     | Number of Patient Admissions |
|-----------------------------------------|------------------------------|
| Alcoholic cirrhosis of the liver        | 1202                         |
| Liver cirrhosis without alcohol mention | 1287                         |
| Hepatic Encephalopathy                  | 666                          |
| Portal Hypertension                     | 1007                         |
| No Liver Disorder                       | 1500                         |



Our Approach achieves the highest F1 score for only 1 of 4 Liver Disorders

Why is our approach not performing as well?

# Limitations

- GPT-4 fails to identify certain concepts in patient notes with high accuracy
  - Could be a potential flaw in our prompting approach – forcing the output to satisfy the template “The patient exhibits \_\_\_\_\_”.
- Our atomic concept features do not contain all possible modifiers
  - For example, “*big*” vs “*small*” *lesion* or “*frequent*” vs “*occasional*” *memory loss*.
- We only consider clinical features but ignore non-clinical features.
  - Non-Clinical Features: Environmental Exposures, Demographics (age, gender), lifestyle (active vs sedentary).
- We only consider clinical features related to dementia.
  - The 254 clinical features related to dementia might not be expressed by a majority of the control group → poor modeling of the negative examples

# Conclusion

- We provide a novel approach for extracting 254 dementia concepts from the Oxford Textbook of Medicine. For example:
  - The patient exhibits memory loss.
  - The patient exhibits ambulatory difficulties.
- We explore different agreement models for converting patient notes to concept vector representations.
- Our Concept Bottleneck approach forces the model to focus on high level clinical concepts, reducing the extent of spurious correlations.
  - Performs well for patient dementia diagnosis
  - Liver disorder performance → highlights limitations and potential areas for future work

# Future Work

- Using LLMs for end-to-end patient diagnosis
  - Unreliable, as model maybe be right for the wrong reasons
  - Blackbox model → almost impossible to find true reasoning of the model
- Breaking the diagnosis pipeline into smaller steps
  - Better interpretability
  - Domain knowledge integration
  - Debugging and troubleshooting
- As a result, my future work will focus on improving diagnostic clinical models using the Concept Bottleneck approach

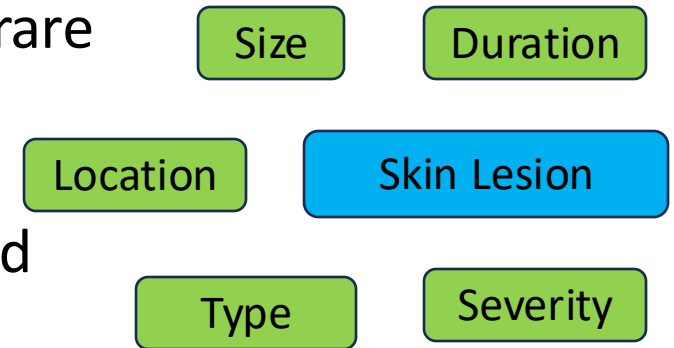
# Future Work: Standardize Concept Features

- *Fever vs Pyrexia*
- *Vomiting vs Emesis*
- *Shortness of breath vs Dyspnea*
- Reduces the dimensionality of the feature space
- UMLS (Unified Medical Language System)
  - Integrates multiple health and biomedical terminologies
    - SNOMED CT (Systematized Nomenclature of Medicine – Clinical Terms)
      - Database for consistently describing symptoms, diagnoses, procedures, etc.
      - 350,000 clinical concepts
    - LOINC (Logical Observation Identifiers Names and Codes)
      - Standardization of test results, vital signs and more
    - ICD-10
      - Global standard for classifying diseases and health conditions



# Future Work: Concept Feature Modifiers

- Create a set of modifiers for each concept feature
  - **Size modifiers:** large, small, tiny, huge
  - **Frequency modifiers:** frequent, occasional, intermittent, rare
  - **Severity modifiers:** mild, moderate, severe, excruciating
  - **Time modifiers:** chronic, acute, recent, persistent
  - **Other clinical modifiers:** bilateral, symmetrical, right-sided



- Concepts with modifiers → concept feature representation
- Resource: SNOWMED CT and the UMLS Metathesaurus
  - Can additionally use LLMs to generate a list of modifiers

# Future Work: Non-Clinical Features

- Create a set of non-clinical concept features
  - **Demographic information:** Age, Gender, Ethnicity
  - **Economic Stability:** Income Level, Employment Status, Housing Stability
  - **Social and Community Context:** Family Support, Social Isolation
  - **Neighborhood:** Housing Quality, Cleanliness, Food Security
  - **Behavioral and Lifestyle Factors:** Physical Activity, Diet, Substance Use, Sleep
- Resource: ICF (International Classification of Functioning, Disability, and Health)

# Future Work: Scaling to More Diseases

- Scaling up our Concept Bottleneck approach to include a universal set of diseases
  - A major hurdle is run time and cost for using LLMs
    - Trends of decreasing costs and higher API calls might make this possible in the future!
    - Newer open-source models like Llama3 show comparable performance to GPT-4
  - Alternatively, improve our Text Embedding approach with GPT-4 to require fewer API calls.
- This will result in better modeling of the control group and a more accurate decision boundary

# Thank you

- Chris Callison-Burch, Mary Regina Boland and Li Shen
- Lyle Ungar, Mark Liberman, Ying Ding
- Lab mates:
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  - Liam Dugan
  - Andrew Zhu
  - Alyssa Hwang
  - Harry Zhang
  - Veronica Qing Lyu
  - Mohammad Sadegh Rasooli
- University of Pennsylvania
- Parents and Partner

